

An Overview of the Theory and Applications of Metasurfaces: The Two-Dimensional Equivalents of Metamaterials

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Abstract

Metamaterials are typically engineered by arranging a set of small scatterers or apertures in a regular array throughout a region of space, thus obtaining some desirable bulk electromagnetic behavior. The desired property is often one that is not normally found naturally (negative refractive index, near-zero index, etc.). Over the past ten years, metamaterials have moved from being simply a theoretical concept to a field with developed and marketed applications. Three-dimensional metamaterials can be extended by arranging electrically small scatterers or holes into a two-dimensional pattern at a surface or interface. This surface version of a metamaterial has been given the name metasurface (the term metafilm has also been employed for certain structures). For many applications, metasurfaces can be used in place of metamaterials. Metasurfaces have the advantage of taking up less physical space than do full three-dimensional metamaterial structures; consequently, metasurfaces offer the possibility of less-lossy structures. In this overview paper, we discuss the theoretical basis by which metasurfaces should be characterized, and discuss their various applications. We will see how metasurfaces are distinguished from conventional frequency-selective surfaces.

Metasurfaces have a wide range of potential applications in electromagnetics (ranging from low microwave to optical frequencies), including: (1) controllable “smart” surfaces, (2) miniaturized cavity resonators, (3) novel wave-guiding structures, (4) angular-independent surfaces, (5) absorbers, (6) biomedical devices, (7) terahertz switches, and (8) fluid-tunable frequency-agile materials, to name only a few. In this review, we will see that the development in recent years of such materials and/or surfaces is bringing us closer to realizing the exciting speculations made over one hundred years ago by the work of Lamb, Schuster, and Pocklington, and later by Mandel'shtam and Veselago.

Keywords: Agile materials; effective media; engineered materials; generalized sheet transition conditions (GSTC); metafilm; metamaterial; metasurface; polarizability densities; surface susceptibility; tunable materials

I. Introduction

In recent years, there has been a great deal of attention devoted to metamaterials [1-10]. Metamaterials are novel synthetic materials engineered to achieve unique properties not normally found in nature. In the context of electromagnetics, early examples of these were artificial dielectrics. Later, we will see that the terms metamaterial and metasurface do not refer to classical periodic structures, such as photonic bandgap (PBG) structures or frequency-selective surfaces (FSS). One particular class of metamaterial that is being studied extensively consists of the so-called double-negative (DNG) materials [1-10] (also known as negative-index materials (NIM), backward-wave (BW) media, or left-handed materials (LHM)). Such materials have the property that their effective permittivity and effective permeability are simultaneously negative in a given frequency band. Another property not normally found in nature that can be achieved with metamaterials is that of near-zero refractive index. In this type of material, either the permittivity or permeability is designed to have its real part close to zero. Materials with unique properties such as these have a wide range of potential applications in electromagnetics at frequencies ranging from the low microwaves to optical, including shielding, low-reflection materials, novel substrates, antennas, electronic switches, “perfect lenses,” and resonators, to name only a few.

Modern metamaterial research activities were stimulated by the theoretical work of Veselago [4], and later by the realization of such structures by Pendry, Smith et al. [5]. However, many researchers in the field today fail to realize that the concept of negative-index materials and their interesting behavior date back much earlier. In fact, these types of materials were anticipated by theory dating back more than a century. Some authors have attributed the first study of such media to Veselago [4] in 1967, but Sivukhin [11] in 1957 briefly examined their properties. Both authors, as well as Malyuzhinets [12] and Silin [13-15], gave credit to much earlier work of L. I. Mandel'shtam [16, 17]. Mandel'shtam himself referred to a 1904 paper of Lamb [18], who may have been the first person to suggest the existence of backward waves (waves with phase and group velocities that are in opposite directions; his examples involved mechanical systems rather than electromagnetic waves). In his 1904 book on optics, Schuster [19] briefly noted Lamb's work, and gave a speculative discussion of its implications for optical refraction should a medium with backward-wave properties ever be found. In 1905, Pocklington [20] showed that a bicycle chain suspended in a certain way could support backward waves, and that a suddenly activated source does indeed produce a wave with a group velocity that is directed away from the source, while the phase velocity moves towards it.

Metamaterials are often engineered by arranging a set of small scatterers (e.g., metallic rings and rods, or spherical magneto-dielectric particles) in a regular array throughout a region of space, thus obtaining some desirable bulk behavior [1-10]. This concept can be extended by judiciously placing electrically small scatterers into a two-dimensional pattern at a surface or interface, as shown in Figure 1. Figure 1a illustrates a general set of scatterers, while Figures 1b-1d show more

specific examples. Figure 1b shows a set of metallic scatterers that generate an electrical response analogous to the magnetic response of the classical split-ring structures. Figure 1c shows an array of spherical particles (of which the three-dimensional metamaterial version was introduced in [7], based on earlier work by Lewin [21], but anticipated almost 100 years earlier by Gans and Happel [22]). Figure 1d shows an array of ceramic cubes. This surface version of a metamaterial was originally given the name metafilm [23, 24], by which is specifically meant a surface distribution of electrically small scatterers. It is important to note that the individual scatterers constituting the metafilm are not necessarily of zero thickness (or are even small compared to the lattice constants): they may be of arbitrary shape, and can have dimensions that are required to be small only in comparison to a wavelength in the surrounding medium. Like that of a metamaterial, the behavior of a metafilm can be characterized by the electric and magnetic polarizabilities of its constituent scatterers. Metafilms have also been referred to in the literature as metasurfaces or single-layer metamaterials. We will attempt to clarify the terminology in Sections 1.1 and 1.2.

For many applications, metasurfaces can be used in place of metamaterials. Metasurfaces have the advantage of taking up less physical space than do full three-dimensional (3D) metamaterial structures; consequently, metasurfaces offer the possibility of less-lossy structures. The use and application of metasurfaces from microwave to optical frequencies has achieved great popularity in recent years [24-82]. In addition to the applications mentioned above for metamaterials, metasurfaces allow for controllable “smart” surfaces, miniaturized cavity resonators, novel wave-guiding structures, compact and wide-angle absorbers, impedance-matching surfaces, and biomedical devices, to name only a few. Some of these will be discussed in more detail later.

1.1 Metasurfaces Compared to Frequency-Selective Surfaces

A few comments are needed on (1) the difference between a metamaterial and a conventional photonic bandgap (PBG) or electromagnetic bandgap (EBG) structure, and, in turn, (2) the difference between a metasurface and a conventional frequency-selective surface (FSS). Consider first three-dimensional (bulk) metamaterials: these ideas will be extended below to metasurfaces. To this end, it is important to understand the electromagnetic behavior of a periodic composite material in various frequency ranges or at different length scales. The behavior of such a composite material can be separated into three distinct regions (see Figure 2), with distinctive behaviors in each region. For a three-dimensional bulk material, Region 1 corresponds to the quasistatic region. This implies low frequencies: more specifically, frequencies at which the wavelength is much larger than the period of the structure (that is, the periodicity of the scatterers that compose the composite medium). These scatterers could have induced or permanent dipole moments, as is the case for atoms or molecules for classical materials. Alternatively, these scatterers could be generic in shape and placed in a host matrix to obtain a man-

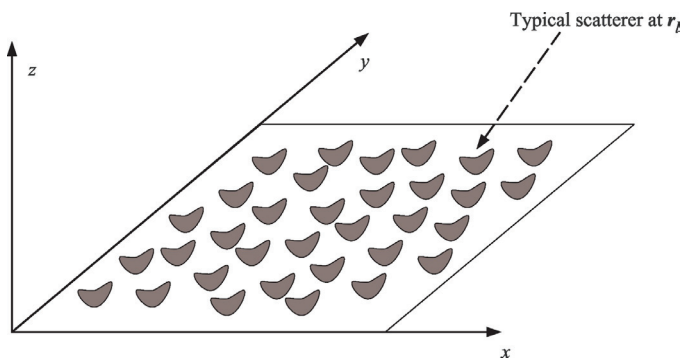


Figure 1a. Illustrations of metasurfaces: a planar array of arbitrarily shaped scatterers.

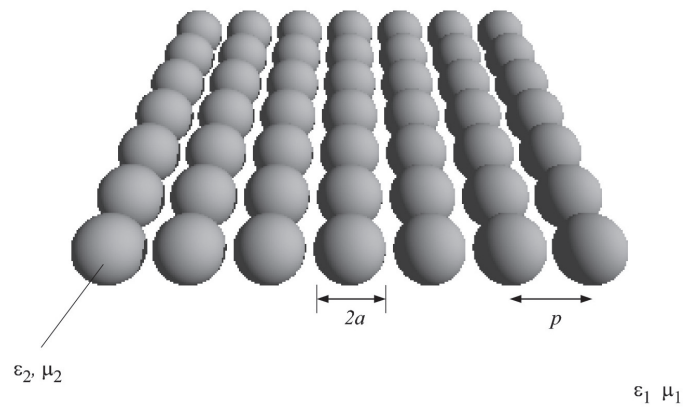


Figure 1c. Illustrations of metasurfaces: planar arrays of spherical particles.

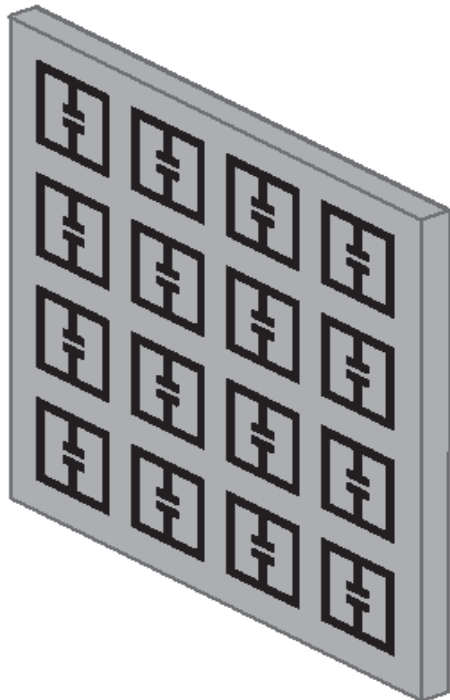
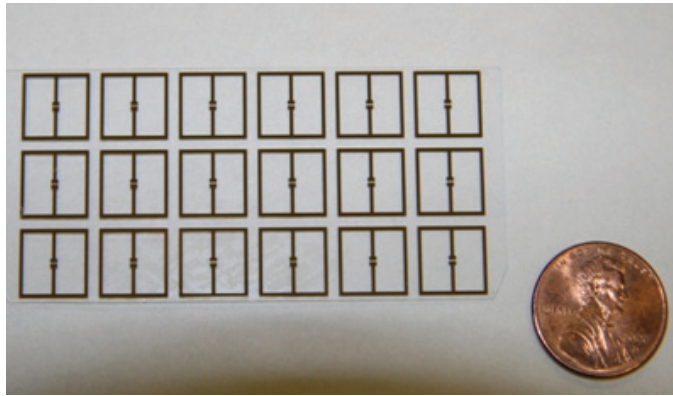


Figure 1b. Illustrations of metasurfaces: planar arrays of metallic scatterers (electrical equivalent of a classical splitting structure).

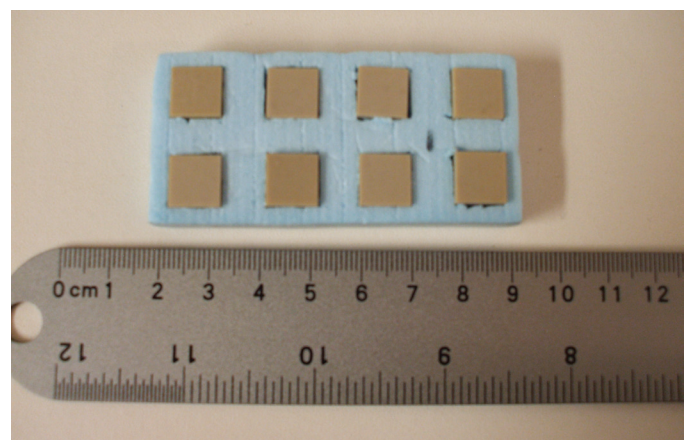
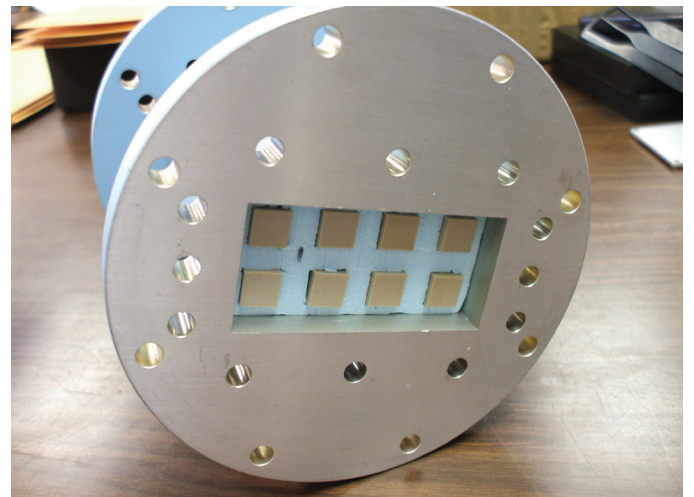


Figure 1d. Illustrations of metasurfaces: an array of ceramic cubes.

made composite material designed to achieve some specified behavior. In this region, classical mixing formulas are used to obtain equivalent effective-material properties (permittivity, ϵ ; permeability, μ ; etc.). The determination of effective-medium properties and the modeling of the electromagnetic response to inclusions embedded in a host material is a problem with a long history going back to Maxwell and Rayleigh.

When the wavelength becomes comparable to or smaller than the period of the structure, the fields no longer “see” the composite as an effective medium. This corresponds to Region 3 in Figure 2. At these frequencies, a more-complicated field behavior exists, and more-elaborate techniques for analyzing the EM field interaction with the composite periodic structure must be used (i.e., full-wave approaches). The classical analytical approach for this is the Floquet-Bloch mode expansion [83-88], in which the fields are expanded into an infinite sum of plane waves propagating in various directions. As the wavelength approaches the period, higher-order Floquet-Bloch modes must be considered. These higher-order modes then interfere with the fundamental wave propagating through the composite, and we call the composite a photonic bandgap or electromagnetic bandgap material in this frequency range. We can say that the material is exhibiting array or lattice dispersion in this case. At certain frequencies, photonic-bandgap and electromagnetic-bandgap structures block the propagation of EM waves through the material. The frequency bands where this occurs are referred to as stop bands. At other frequencies, the periodic structure allows energy to propagate through the structure: these frequency bands are referred to as pass bands. Phenomena such as Bragg scattering are associated with this frequency region, and are the basis of many practical applications [1, 88].

Region 2 in Figure 2 corresponds to a region where the period of the structure is still small compared to a wavelength, but the individual scatterers are designed in such a manner (either via their shape or their constitutive properties) that the scatterers themselves can resonate. When this occurs, a new class of manmade materials (metamaterials) is realized, making possible a broad range of unique behavior not commonly found in nature (double-negative or near-zero index materials, for example). Region 2, where the scatters resonate (but not the lattice, as is the case in Region 3), is where we call our medium a metamaterial. The medium is dispersive here, but we may still characterize its electromagnetic behavior with the effective parameters ϵ and μ .

Similar behavior (three regions of behavior) will occur for scatterers in a two-dimensional array (a metasurface). For a two-dimensional lattice of scatterers, Region 1 corresponds to classical thin-film materials, while Region 3 corresponds to resonances associated with the periodicity of the scatterers. The conventional frequency-selective surface [87] falls into this third region. On the other hand, when we talk about a metasurface, we are referring to an array of scatterers that falls into Region 2. Resonances of the surface may be associated with the resonances of the scatterers, but not with the periodicity of the array. Ordinary frequency-selective surfaces are

sometimes operated in this regime, but the distinction between this type of operation and that of Region 3 has not usually been made.

We need to emphasize that this second region in Figure 2 may not always occur. The scatterers need to be properly designed, such that the scatterers’ resonances occur at a frequency well below that where the next higher-order Floquet-Bloch mode can propagate. For example, if the bulk properties and the radius of a spherical-particle composite (see [7]) are too small, or if the sizes or shapes of the scatterers used in the material are not properly chosen, the scatterers’ resonances would be pushed toward the Floquet-Bloch-mode region, and a double-negative material could not be realized [7]. The scatterers’ resonance region would be absorbed into the Floquet-Bloch-mode region, and an effective-medium model would not adequately describe the behavior of the composite material.

In summary, in Regions 1 and 2 of Figure 2 the electromagnetic field behaves as if the composite material is some type of effective medium. In Region 1 (the classical mixing-theory region), the effective material properties are not frequency dependent (except insofar as the constituent bulk properties have a frequency dependence). In Region 2 (the scatterers’ resonant region), the material behaves as an effective medium and has an inherent frequency dependence exhibited in the effective material properties. In this region, it is possible to achieve three-dimensional double-negative materials and other interesting resonant behaviors (e.g., index near zero). In the last region (Region 3 in Figure 2), the electromagnetic field’s interaction with the periodic structure is very involved. We may no longer think of the composite material as behaving like an effective medium. When the wavelength approaches the period, higher-order Floquet-Bloch modes must be considered, and one typically does not refer to these materials as metamaterials or metasurfaces in this region.

1.2 Types of Metasurfaces

We will call any periodic two-dimensional structure the thickness and periodicity of which are small compared to a wavelength in the surrounding media a metasurface. Within this general designation, we identify two important subclasses.

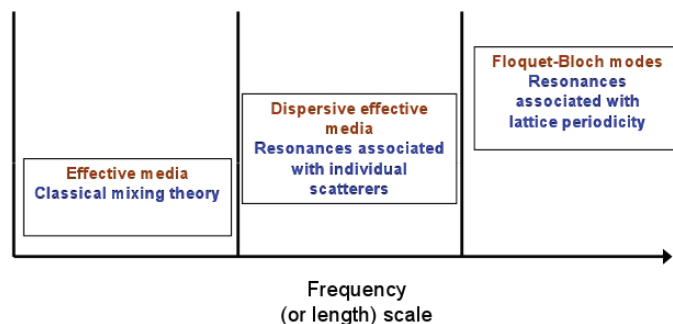


Figure 2. Three characteristic regions of a composite material.

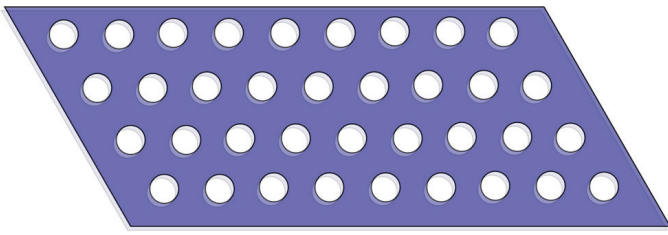


Figure 3. A metascreen.

Metasurfaces that have a “cermet” topology – which refers to an array of isolated (non-touching) scatterers, as shown in Figure 1a – are called metafilms, a term coined in [23] for such surfaces. Metasurfaces with a “fishnet” structure, as shown in Figure 3, will be called metascreens. These are characterized by periodically spaced apertures in an otherwise relatively impenetrable surface. Other kinds of metasurfaces lie somewhere between these two extremes. For example, a grating of parallel conducting wires behaves like a metafilm in the direction perpendicular to the wires’ axes, but like a metascreen in the direction along these axes.

1.3 Modeling a Metasurface

The traditional and most convenient method by which to model metamaterials is with effective-medium theory. In this approach, some type of averaging is performed on the electric and magnetic fields over a given period cell composing the metamaterial. From these averages, the effective permittivity and permeability (and hence, the refractive index) of the metamaterials are determined. It should be emphasized that the averaging (also called homogenization) is valid only when the wavelength is “large enough” compared to the lattice constant of the period cell. While period-cell averaging for the fields is the correct method for defining effective material properties (see [61, 89-93] for this type of homogenization averaging), many researchers have in practice used an approach based upon the reflection and transmission coefficients of a metamaterial sample of some defined thickness. The Nicolson-Ross-Weir (NRW) approach [94, 95] or a variant of it is then used to obtain the effective material properties of the bulk metamaterials. Note that the standard NRW approach must be modified when negative material properties exist. Typically, the choice of the sign of a square root is made unambiguous by ensuring positive power flow in the direction of propagation [96-98]. In some situations, local effects near the boundaries of the sample must also be taken into account [99]. If used properly, the effective-medium approach can be a self-consistent and unique method for characterizing a metamaterial.

Attempts to use a similar bulk-parameter analysis for metasurfaces have been less successful (see [100] for a detailed discussion on this point). Indeed, some previous metasurface studies have modeled the film as a single-layer metamaterial in which effective bulk material properties of the metasurface are obtained by arbitrarily introducing a nonzero thickness parameter into the analysis. Several problems arise from the physically artificial character of this parameter: such a bulk-

property characterization of a metasurface is incorrect at a fundamental level. To the extent that classical algorithms for bulk-parameter extraction give results that depend on sample size, we must conclude that some localized effect is occurring near the boundary of the sample, analogous to the effect of cutoff modes near the junction between two different waveguides.

To illustrate the difficulties that arise, let us replace the metasurface in Figure 1 by an equivalent material layer of thickness d (see Figure 4). The effective material properties of this equivalent layer are determined from the same modified NRW approach used to analyze the bulk three-dimensional metamaterial as discussed above. The problem is that effective bulk material properties are not uniquely defined for a metasurface. While the geometry of the scatterers and the lattice constant, p , are uniquely defined, the thickness of the equivalent layer, d , is not. In fact, one could choose d to be any value, chosen within reasonable limits. Since d is not uniquely defined, the effective material properties obtained in this way suffer from the same ambiguity, and hence are not uniquely defined. Therefore, contrary to what has been asserted elsewhere in the literature [29-34], characterizing the metasurface as an effective medium in this way is inappropriate and leads to misinterpretations. In [30], Smith et al. addressed this point by asserting that the quantity “ ϵd ” is most likely constant. The connection between the surface susceptibilities and bulk effective material properties was also discussed in [101], where it was argued that for a metasurface, one cannot define bulk parameters (ϵ and μ) that are independent of d . As such, it is the product of d and a bulk parameter (ϵ or μ) that must be fixed and unique. This results in an arbitrariness in the choices of d , ϵ , and μ . With that said, one can use these retrieved effective material properties as long as one also assumes that the slab thickness is held constant. One cannot necessarily use the same effective material properties for a slab of a different thickness, let alone for any other non-planar shape of the effective medium. As long as we are willing to use these effective material properties with a set thickness, they can be used without any loss of generality. On the other hand, if one changes the thickness, but keeps the effective properties the same, then the results would be meaningless.

This point is emphasized in Figures 5 and 6. Figure 5 shows the real parts of the effective permittivity for an array of spherical particles obtained in this manner for different values of d (where d is the effective thickness). Figure 6 shows the results for the refractive index for different thicknesses for an array of metallic scatterers. The effective properties shown in Figures 5 and 6 are seen to be functions of the chosen thickness, d , which illustrates that these effective bulk properties are not an intrinsic property of the metasurface. We should also add that the effective bulk properties of a metafilm/metamaterial

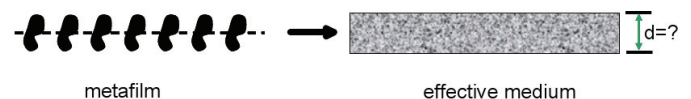


Figure 4. Representing a metafilm as a bulk effective medium of thickness d .

extracted in this way can display non-physical behaviors, as well. For example, causality (as manifested in the Kramers-Kronig relations) may be violated, or a passive material may exhibit a bulk permittivity or permeability with gain (a positive imaginary part for the $e^{j\omega t}$ time dependence assumed in this paper).

In [100], it was shown that the surface susceptibilities are unique properties of a metafilm, and as such, serve as the most-appropriate manner of characterizing this type of metasurface. Physically speaking, the metasurface acts like an infinitesimal sheet: one that causes a phase shift (and possibly a change in amplitude) in wave fields that interact with it. Scattering by such a metasurface is best characterized by generalized sheet-transition conditions (GSTCs) [23], in contrast to the effective-medium description used for a metamaterial. The coefficients appearing in the generalized sheet-transition conditions for any given metasurface are all that are required to model the metasurface's interaction with an electromagnetic field.

For a metafilm, it was shown in [23] that the generalized sheet-transition conditions relate to the electromagnetic fields on both sides of the metasurface (see Figure 1a):

$$\begin{aligned} \mathbf{a}_z \times \mathbf{H}|_{z=0^-}^{0+} &= j\omega \tilde{\chi}_{ES} \cdot \mathbf{E}_{t,av}|_{z=0} - \mathbf{a}_z \times \nabla_t \left[\chi_{MS}^{zz} H_{z,av} \right]_{z=0} \\ \mathbf{E}|_{z=0^-}^{0+} \times \mathbf{a}_z &= j\omega \tilde{\chi}_{MS} \cdot \mathbf{H}_{t,av}|_{z=0} + \mathbf{a}_z \times \nabla_t \left[\chi_{ES}^{zz} E_{z,av} \right]_{z=0} \end{aligned} \quad (1)$$

where the subscript “av” represents the average of the field on either side of the metasurface, the subscript t refers to compo-

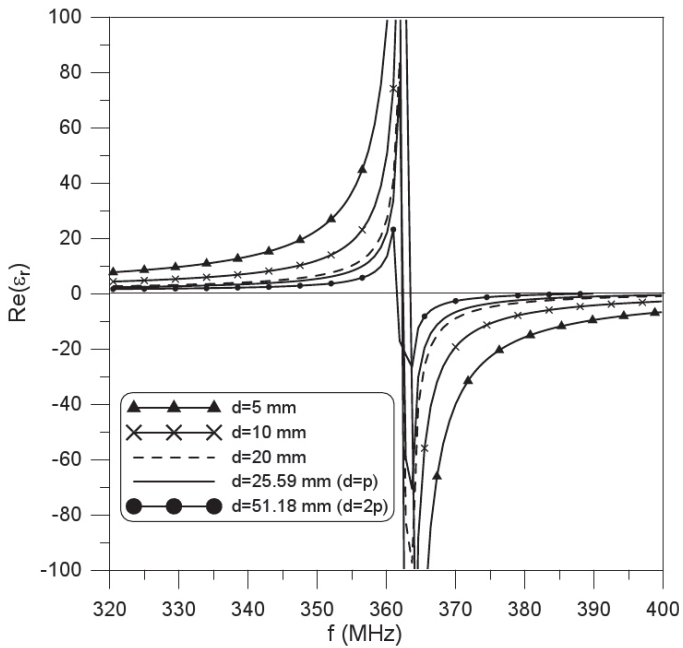


Figure 5. The effective permittivity of a metafilm modeled as an effective medium for different layer thicknesses. The metafilm in this figure was composed of spherical particles as seen in Figure 1c.

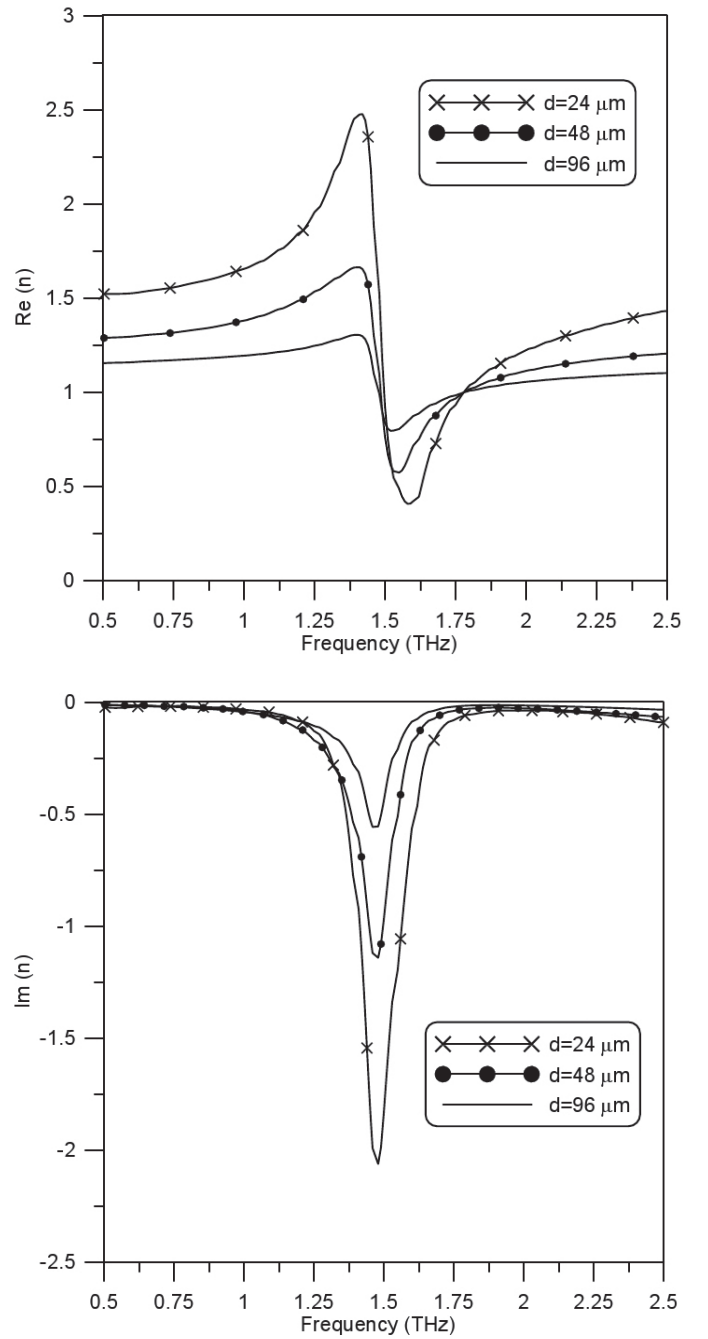


Figure 6. The real (top) and imaginary (bottom) parts of the refractive index of a metafilm modeled as an effective medium for different layer thicknesses. The metafilm in this figure was composed of square metallic scatterers as in Figure 1b.

nents transverse to z , and $\mathbf{a}_z$ is the unit vector. The parameters $\tilde{\chi}_{ES}$ and $\tilde{\chi}_{MS}$ are the dyadic surface electric and magnetic susceptibilities. These have units of meters, and are related to the electric and magnetic polarizability densities of the scatterers per unit area. These dyadics vanish when the scatterers are absent, and Equation (1) reduces to the ordinary conditions of continuity of the tangential components of $\mathbf{E}$ and $\mathbf{H}$. For simplicity, we consider here only the case where the scatterers and lattice have sufficient symmetry such that the surface susceptibility dyadics are diagonal:

$$\tilde{\chi}_{ES} = \chi_{ES}^{xx} \mathbf{a}_x \mathbf{a}_x + \chi_{ES}^{yy} \mathbf{a}_y \mathbf{a}_y + \chi_{ES}^{zz} \mathbf{a}_z \mathbf{a}_z, \quad (2)$$

$$\tilde{\chi}_{MS} = \chi_{MS}^{xx} \mathbf{a}_x \mathbf{a}_x + \chi_{MS}^{yy} \mathbf{a}_y \mathbf{a}_y + \chi_{MS}^{zz} \mathbf{a}_z \mathbf{a}_z.$$

In this paper and in [101], we have chosen a sign convention for χ_{MS} such that there is a minus sign in front of the second term on the right-hand side (RHS) for the jump in the $\mathbf{H}$ field, and a plus sign in front of the first term on the right-hand side for the jump in the $\mathbf{E}$ field (see Equation (1)). Note that this is a different sign convention than that used in [100]. We have chosen this new sign convention because it ensures that $\text{Re}(\chi_{MS}) > 0$ when the surface magnetization density is in the same direction as $\mathbf{H}$ (paramagnetic material), which is consistent with the definition of bulk susceptibilities of classical magnetic materials.

The generalized sheet-transition conditions for the more-general case of non-symmetric, bi-isotropic, and bianisotropic surface susceptibility dyadics are possible, and the theory for these has yet to be fully developed. We should emphasize that the generalized sheet-transition conditions of Equation (1) are appropriate only for metafilms. Metasurfaces with other structures will require a different form of the generalized sheet-transition conditions. For example, metascreens consisting of periodic isolated apertures in a zero-thickness perfect conductor obey the Kontorovich form of generalized sheet-transition conditions [83], which can be written in the form

$$\mathbf{E}|_{z=0} \times \mathbf{a}_z = j\omega\mu\tilde{\pi}_{MS} \cdot \mathbf{H}|_{z=0}^{0+} + \frac{1}{\epsilon} \mathbf{a}_z \times \nabla_t \left[\pi_{ES}^{zz} D_z \right]_{z=0}^{0+} \quad (3)$$

where the tangential $\mathbf{E}$ field and the normal $\mathbf{B}$ field are continuous across the metascreen. Here, the dyadics $\tilde{\pi}_{ES}$ and $\tilde{\pi}_{MS}$ are what we call the electric and magnetic porosities of the metascreen, and, like the surface susceptibilities of a metafilm, these have dimensions of length. When the apertures are absent and we have only the perfectly conducting plane, the electric and magnetic porosity dyadics vanish, and Equation (3) reduces to the vanishing of tangential $\mathbf{E}$, as expected. The most general form of metascreen boundary conditions has not yet been determined, and its development is the subject of ongoing work. The same is true for metasurfaces such as wire gratings, which exhibit features of both metafilms and metascreens (cf. the generalized sheet-transition conditions for thin-wire gratings in [102] and [103]).

The generalized sheet-transition conditions can also be cast in the form of impedance-type boundary conditions [83, 104-107]. For plane-wave fields the variation of which parallel to a metafilm is of the form

$$e^{-j\mathbf{k} \cdot \mathbf{r}_t} \text{ (where } \mathbf{k} = k_x \mathbf{a}_x + k_y \mathbf{a}_y \text{ and } \mathbf{r}_t = x \mathbf{a}_x + y \mathbf{a}_y \text{),} \quad (4)$$

we can use Maxwell's equations to write Equation (1) as

$$\mathbf{a}_z \times \mathbf{H}|_{z=0}^{0+} = \tilde{Y}_{ES} \cdot \mathbf{E}_{t,av}|_{z=0}, \quad (5)$$

$$\mathbf{E}|_{z=0}^{0+} \times \mathbf{a}_z = \tilde{Z}_{MS} \cdot \mathbf{H}_{t,av}|_{z=0},$$

in which the spatially dispersive ($\mathbf{k}$ -dependent) surface transfer admittance and transfer impedance are given by

$$\tilde{Y}_{ES} = j\omega\tilde{\chi}_{ES} + \frac{j\chi_{MS}^{zz}}{\omega\mu} (\mathbf{a}_z \times \mathbf{k})(\mathbf{a}_z \times \mathbf{k}) \quad (6)$$

$$\tilde{Z}_{MS} = j\omega\tilde{\chi}_{MS} + \frac{j\chi_{ES}^{zz}}{\omega\epsilon} (\mathbf{a}_z \times \mathbf{k})(\mathbf{a}_z \times \mathbf{k}).$$

Such boundary conditions can also be interpreted as equivalent transmission-line circuits [105].

These generalized sheet-transition conditions allow a metasurface to be modeled by a boundary condition that is applied across an infinitely thin *equivalent* surface. The details of the surface geometry are incorporated into this boundary condition through the surface susceptibilities or porosities. The fields that appear in the generalized sheet-transition conditions are “macroscopic” fields, in the sense that they do not exhibit variations on a length scale comparable to scatterer or aperture dimensions or the lattice spacing, but only on larger scales such as the wavelength in the surrounding medium. It should be noted that the generalized sheet-transition conditions as written here do contain (in a first approximation) spatial-dispersion effects via the presence of the surface gradients in Equations (1) and (3). A more-accurate account of these effects can be given if the surface susceptibilities and porosities are allowed to be functions of the vector wavenumber (i.e., of the direction of propagation of the waves).

If the fine structure of the spatial field variation is not required, these generalized sheet-transition conditions along with Maxwell's equations are all that are required to analyze the interaction of the fields with the metasurface. The surface susceptibility and porosity dyadics that appear explicitly in the generalized sheet-transition conditions are uniquely defined (unlike in a bulk effective-parameter model), and, as such, represent the physical quantities that uniquely characterize the metasurface. Retrieval techniques based on this model are described in the next section.

1.4 Characterization of Metasurfaces

As discussed in the last section, the bulk effective permittivity and permeability (as well as the refractive index) of a metasurface are not uniquely defined. In this section, we present an approach for the unique characterization of a metafilm based on an inversion of its reflection and transmission coefficients to obtain the surface susceptibilities, details of which were discussed in [100] and [101].

The reflection (R) and transmission (T) coefficients of a metafilm for both TE and TM polarized plane waves (see Figure 7) were derived in [24]. The reflection coefficient, R , and the transmission coefficient, T , for a TE-polarized plane wave are given by

$$R_{\text{TE}}(\theta) = \frac{-j \frac{k_0}{2 \cos \theta} (\chi_{ES}^{yy} + \chi_{MS}^{zz} \sin^2 \theta - \chi_{MS}^{xx} \cos^2 \theta)}{D_1},$$

$$T_{\text{TE}}(\theta) = \frac{1 + \left(\frac{k_0}{2}\right)^2 \chi_{MS}^{xx} (\chi_{ES}^{yy} + \chi_{MS}^{zz} \sin^2 \theta)}{D_1}, \quad (7)$$

$$D_1 \equiv 1 - \left(\frac{k_0}{2}\right)^2 \chi_{MS}^{xx} (\chi_{ES}^{yy} + \chi_{MS}^{zz} \sin^2 \theta) + j \frac{k_0}{2 \cos \theta} (\chi_{ES}^{yy} + \chi_{MS}^{xx} \cos^2 \theta + \chi_{MS}^{zz} \sin^2 \theta),$$

and for a TM polarized plane wave by

$$R_{\text{TM}}(\theta) = \frac{-j \frac{k_0}{2 \cos \theta} (\chi_{ES}^{xx} \cos^2 \theta - \chi_{MS}^{yy} - \chi_{ES}^{zz} \sin^2 \theta)}{D_2}$$

$$T_{\text{TM}}(\theta) = \frac{1 + \left(\frac{k_0}{2}\right)^2 \chi_{ES}^{xx} (\chi_{MS}^{yy} + \chi_{ES}^{zz} \sin^2 \theta)}{D_2}, \quad (8)$$

$$D_2 \equiv 1 - \left(\frac{k_0}{2}\right)^2 \chi_{ES}^{xx} (\chi_{MS}^{yy} + \chi_{ES}^{zz} \sin^2 \theta) + j \frac{k_0}{2 \cos \theta} (\chi_{MS}^{yy} + \chi_{ES}^{xx} \cos^2 \theta + \chi_{ES}^{zz} \sin^2 \theta),$$

where k_0 is the free-space wavenumber. These reflection and transmission coefficients apply to a metafilm the constituent scatterers of which have sufficient symmetry that the surface susceptibility dyadics are diagonal. The more-general case of non-symmetric, bi-isotropic surface susceptibility dyadics could also be handled in a similar way, as well as that of a metascreen or more-general metasurface. These cases have yet to be developed in full.

Once the reflection and transmission coefficients are obtained (either from measurements or from numerical calculations), the surface susceptibilities can be determined. For the present case, two different sets of R and T are required (for example, one at normal incidence and one at oblique incidence) for each polarization. For a TE-polarized wave, the three unknown surface susceptibilities are determined from the following:

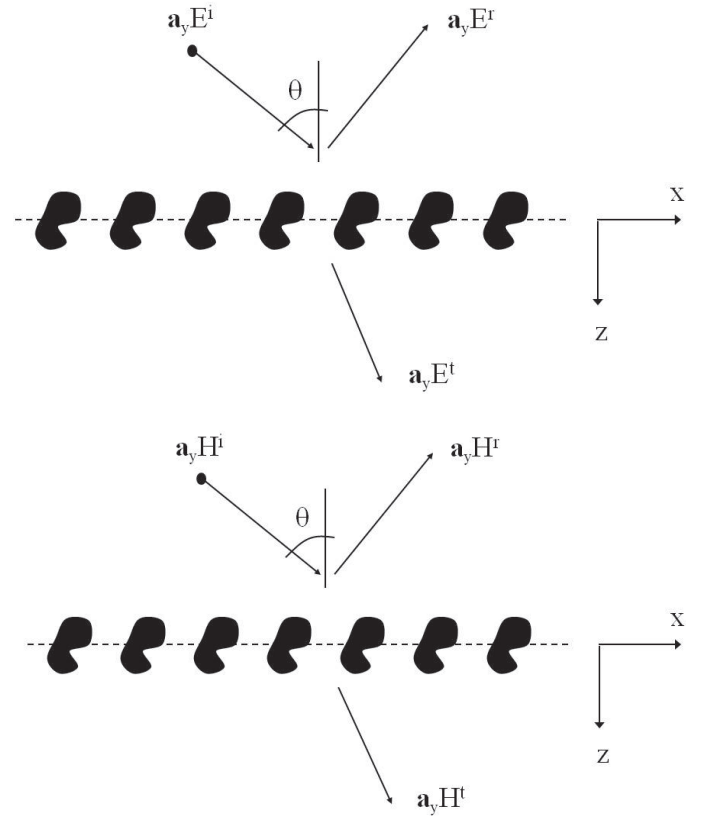


Figure 7. A plane wave incident onto a metafilm: (top) TE polarization, and (bottom) TM polarization.

$$\chi_{MS}^{xx} = \frac{2j}{k_0} \frac{R_{\text{TE}}(0) - T_{\text{TE}}(0) + 1}{R_{\text{TE}}(0) - T_{\text{TE}}(0) - 1}, \quad (9)$$

$$\chi_{ES}^{yy} = \frac{2j}{k_0} \frac{R_{\text{TE}}(0) + T_{\text{TE}}(0) - 1}{R_{\text{TE}}(0) + T_{\text{TE}}(0) + 1},$$

and

$$\chi_{MS}^{zz} = -\frac{\chi_{ES}^{yy}}{\sin^2(\theta)} + \frac{2j \cos(\theta)}{k_0 \sin^2(\theta)} \frac{R_{\text{TE}}(\theta) + T_{\text{TE}}(\theta) - 1}{R_{\text{TE}}(\theta) + T_{\text{TE}}(\theta) + 1}, \quad (10)$$

where $R(0)$ and $T(0)$ are the reflection and transmission coefficients at normal incidence, and $R(\theta)$ and $T(\theta)$ are the reflection and transmission coefficients at some oblique incidence angle, θ . For a TM-polarized wave, the three unknown surface susceptibilities are determined by

$$\chi_{ES}^{xx} = \frac{2j}{k_0} \frac{R_{\text{TM}}(0) + T_{\text{TM}}(0) - 1}{R_{\text{TM}}(0) + T_{\text{TM}}(0) + 1}, \quad (11)$$

$$\chi_{MS}^{yy} = \frac{2j}{k_0} \frac{R_{\text{TM}}(0) - T_{\text{TM}}(0) + 1}{R_{\text{TM}}(0) - T_{\text{TM}}(0) - 1},$$

and

$$\chi_{ES}^{zz} = -\frac{\chi_{MS}^{yy}}{\sin^2(\theta)} + \frac{2j \cos(\theta)}{k_0 \sin^2(\theta)} \frac{T_{TM}(\theta) - 1 - R_{TM}(\theta)}{T_{TM}(\theta) + 1 - R_{TM}(\theta)}. \quad (12)$$

The sign errors that were printed in [100] are corrected in the expressions given above. Corrections were also given in [101].

This approach can be applied equally well either to numerically or experimentally determined values of R and T . However, because of the difficulty of separating the incident and reflected components at normal incidence in measurements, it may be more beneficial to rewrite these equations for two arbitrary incidence angles that both differ from zero, as discussed in [100]. The interesting thing to notice is that this retrieval approach for characterizing a metafilm by its surface susceptibilities does not pose the problem of choosing the appropriate sign of a square root, nor require the assumption of an arbitrary layer thickness, as is the case when trying to infer the effective properties for a bulk metamaterial model.

In order to provide a validation of these expressions for retrieving the surface susceptibilities of a metafilm, we consider the example of a metafilm consisting of a square array of spherical particles, as in Figure 1c. In this example, we chose $a = 10$ mm, $p = 25.59$ mm, $\epsilon_r = 2$, $\mu_r = 900$, and $\tan \delta = 0.04$. This structure is appealing because the polarizabilities of the spherical particles are known analytically, and hence the surface susceptibilities can be calculated analytically by use of expressions given in [100]. As suggested above, there are various possible sets of the R and T that could be used to determine the unknown susceptibilities, and several approaches were discussed in [100].

Figure 8 shows the real and imaginary parts of χ_{ES}^{yy} and χ_{ES}^{zz} . These results were obtained from numerically simulated values of R and T for both polarizations at a 30° incidence angle. The metafilm used in this example was a square array of lossy spherical particles, and thus had symmetric transverse surface susceptibilities (or $\chi_{ES}^{xx} = \chi_{ES}^{yy}$ and $\chi_{MS}^{xx} = \chi_{MS}^{yy}$). Also shown in the figure are the analytical results, details of which were given in [100]. From this comparison, we saw that the surface susceptibilities obtained from the retrieval approach were virtually identical to those obtained from the approximate analytical expressions.

The example given here and others presented in [100] illustrate the validity of this retrieval approach. More importantly, these results illustrate that the surface susceptibilities are unique properties of a metafilm, and as such serve as the most appropriate way to characterize them.

Metafilms can also be used as a means of extracting *bulk* properties of three-dimensional metamaterials [108]. In this technique, plane-wave reflection and transmission coefficients from a metafilm are used to extract the surface susceptibilities, as above. From [23] (or more properly, a refinement of the

formulas therein), we can then obtain the polarizabilities of the individual scatterers in the metafilm. These are finally inserted into the Clausius-Mossottii relations to obtain the effective permittivity and permeability of a metamaterial realized by stacking these metafilms in a third dimension. This technique is a useful alternative to standard parameter-extraction methods when transmission through a sample of many layers is very small.

In the remainder of the paper, we will discuss various applications of metasurfaces that have emerged over the last few years.

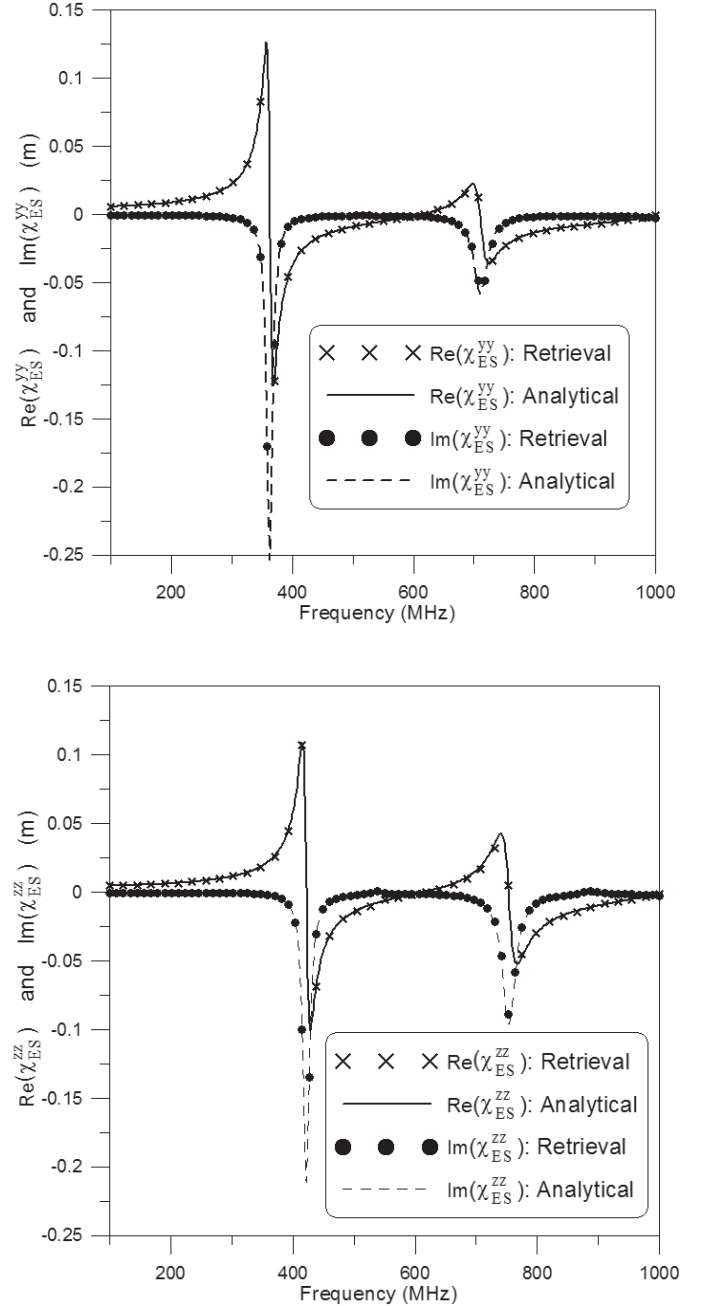


Figure 8. The surface susceptibilities for a metafilm composed of spherical particles.

2. Controllable Surfaces

Given a generic metasurface, one could use one of a number of the commercial computational codes to analyze the interaction of an electromagnetic field with a metasurface. However, as we have seen, the generalized sheet-transition conditions allow us to obtain closed-form expressions for the reflection and transmission coefficients of a plane wave. For a metafilm, they take the form of Equations (7) and (8). These closed-form relationships are expressed in terms of the surface susceptibilities of the metafilm, which are related to the scatterers that make it up. The advantage of the closed-form expressions is that one can get some physical insight into the relationship of the surface susceptibilities to the reflection and transmission behavior. Expressions (7) and (8) illustrate that if the surface susceptibility (or the electric and magnetic polarizability of the individual scatterers) can be changed, it is possible to control the reflection and transmission behavior of the surface.

As examples, from Equations (7) and (8), one can write down the relationship between the electric and magnetic surface susceptibilities needed to have either total reflection or total transmission [24]. For total reflection, the following conditions must be satisfied:

$$k_0^2 \chi_{MS}^{xx} (\chi_{ES}^{yy} + \chi_{MS}^{zz} \sin^2 \theta) = -4 \text{ for TE,} \quad (13)$$

$$k_0^2 \chi_{ES}^{xx} (\chi_{MS}^{yy} + \chi_{ES}^{zz} \sin^2 \theta) = -4 \text{ for TM,}$$

while for total transmission, the required conditions are

$$\begin{aligned} \chi_{ES}^{yy} - \chi_{MS}^{xx} \cos^2 \theta + \chi_{MS}^{zz} \sin^2 \theta &= 0, \\ \chi_{MS}^{yy} - \chi_{ES}^{xx} \cos^2 \theta + \chi_{ES}^{zz} \sin^2 \theta &= 0. \end{aligned} \quad (14)$$

In principle, control of the reflection and transmission behavior can be done in a number of ways, e.g., (1) by changing the electrical or magnetic properties of the scatterers, or (2) by changing the properties (either the material properties or the geometry) of the substrate on which the scatterers lie. For example, for a metafilm made up of spherical magnetic particles, Figure 9 shows R as a function of the permeability of the scatterers [24]. From this figure, we see that the metafilm goes from total reflection to total transmission as the permeability of the spherical particles is varied. Such a controllable surface has been realized by using a metafilm of spherical YIG particles, and controlling the surface behavior with a dc magnetic bias field [25]. Figure 10 shows the transmission behavior of such a metafilm as a function of the external dc magnetic field. Other approaches have been used to control the surface behavior of a metasurface, and a significant amount of research is ongoing in the area of dynamically controllable metasurfaces, as we will discuss in Sections 7 and 9.

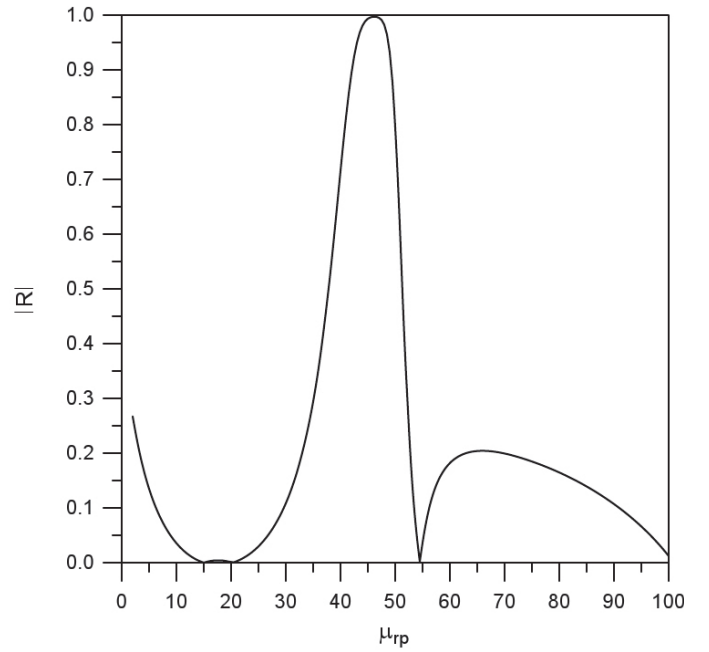


Figure 9. The reflection coefficient as a function of the permeability, μ_{rp} , of the spherical particles comprising the metasurface.

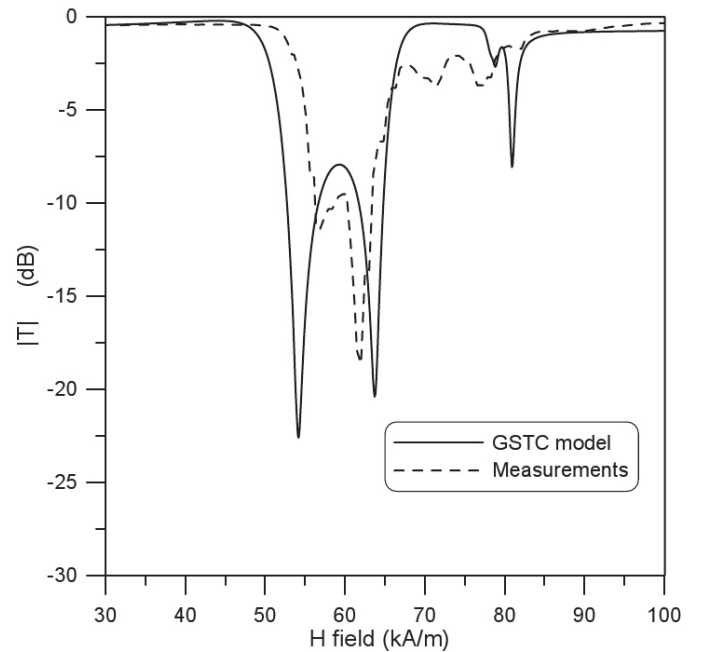


Figure 10. Controlling the transmission properties of a surface with an external dc magnetic field.

3. Anisotropic Metasurfaces: Angularly Independent Behavior, Absorbers, and Impedance-Matching Surfaces

The use of anisotropic scatterers to make up a metafilm can be used to achieve a number of useful results not possible with isotropic metasurfaces. One of the more interesting aspects of bulk metamaterials is the ability to engineer anisotropic behavior into them. In fact, this is the idea behind so-called “cloaking” materials, which has generated so much media interest. The cloaking concept is to design a specific anisotropic behavior into a metamaterial in such a way as to cause electromagnetic energy to propagate or bend around an object that is covered with it [65, 66]. If the anisotropic materials are correctly designed, the coated object will neither scatter nor absorb energy, and hence it appears to the electromagnetic (EM) fields as if the object is not present. One could in principle thus achieve an object that appears invisible to the EM energy in any desired frequency band, from RF to optical. We might add that due to the inherent properties of materials and structures, there are technical hurdles to overcome (some may appear to be impossible) in order to obtain the “holy grail” of broadband cloaking metamaterials. Only narrowband, single-polarization examples have been experimentally demonstrated to date.

This concept could be extended to metasurfaces, as well. In such a design, the scatterers composing a metasurface would be chosen and varied across the surface in order to achieve a desired surface behavior. For example, it is in principle possible to design a metasurface so as to focus an EM plane wave to a desired region in space, much like a focusing antenna array. If the scatterers of the metasurface were designed in a manner such that they could be changed at will, one could as a result have a metasurface capable of changing the direction and frequency where the energy is focused, i.e., a frequency- and space-agile surface. These concepts are currently being investigated.

Another example deals with how to obtain (at least in some parameter range) angularly independent behavior of the reflection and transmission [60]. For a given incidence angle, Equation (13) gives the relationship between the electric and magnetic surface susceptibilities required for total reflection. Equation (13) suggests a mechanism by which one could achieve approximately the angular independence of such total reflection. For instance, in the TE case, when $\chi_{ES}^{yy} \gg \chi_{MS}^{zz}$, the angular dependence becomes less significant, because the first term in the parentheses of Equation (13) dominates over the second angularly dependent term. Similarly, for the TM case, when $\chi_{MS}^{yy} \gg \chi_{ES}^{zz}$, the angular dependence likewise becomes weaker. These conditions could occur, for example, at certain resonance frequencies of the scatterers that constitute a metafilm. When the indicated terms are dominant, the total reflection conditions, Equation (13), become

$$\chi_{MS}^{xx} \chi_{ES}^{yy} = -\frac{4}{k_0^2} \text{ for TE,} \quad (15)$$

$$\chi_{ES}^{xx} \chi_{MS}^{yy} = -\frac{4}{k_0^2} \text{ for TM.}$$

If a metasurface is thus designed such that the transverse components of the surface susceptibilities are highly resonant as compared to the normal components, then the metasurface may exhibit angle-independent behavior. This behavior is illustrated in Figure 11, where we have plotted the reflection coefficient of a metasurface composed of the metallic structures shown in Figure 1b [60]. In this figure, we see that the surface exhibits a nearly angle-independent behavior up to incidence angles as large as 60°.

One can extend this concept to many other types of structures. For example, it can be shown that highly anisotropic slabs of materials will exhibit some type of angle-independent behavior, as well. This can be seen by examining the reflection properties of an inhomogeneous anisotropic slab, as discussed in [61] (see Equations (31)-(38) therein). Additionally, the mathematically constructed perfectly matched layer (PML) [62] that was introduced into computational electromagnetics for reducing non-physical reflections at radiation boundaries also requires such angular independence. Such perfectly matched layers may be physically realizable by use of the concept of metasurfaces. The idea of designing a metasurface structure to use as a perfectly matched layer in computational codes is under investigation.

This mechanism offers a unique way to match lossy materials for the purpose of developing compact electromagnetic absorbers, and to make possible unique designs of impedance-matching surfaces. Compact absorbers based on such ideas have recently begun to appear in the literature [53-58]. These

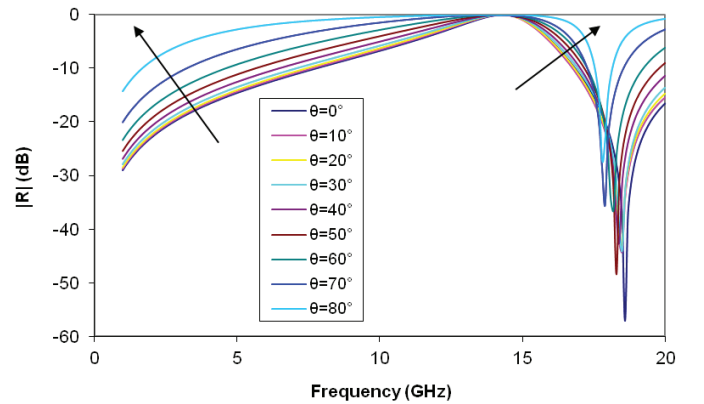


Figure 11. The reflection coefficient for a TE-polarized incident plane wave for an electrical resonator metasurface structure.

structures typically are designed with a metasurface on the front side of a lossy substrate backed by a metal plate (see the second images in Figure 1b). These structures may be narrowband, but are quite compact in size. An additional benefit of using the metasurface is the fact that these structures show substantial angular independence, which has been demonstrated both numerically and experimentally [53, 55, 57]. The physical explanation of this angularly independent behavior is discussed above and in [60].

Various groups have also investigated the use of metasurfaces to develop impedance-matching surfaces. An array of metallic structures similar to the one shown in Figure 12a was used to design and fabricate a wide-angle impedance-matching surface [59]. This structure had highly anisotropic properties, which as we saw above can enable angularly independent behavior. The reflection properties of such a thin anisotropic metasurface are shown in Figure 12b.

4. Resonator Size Reduction

Engheta [63, 64] demonstrated that the $\lambda/2$ classical lower bound on the size of a resonant structure can be reduced if a cavity is partially filled with a negative-index material. Extending this idea, it was shown in [35, 36] that the same thing could be accomplished with a metasurface. The advantage of a metasurface is that because it requires less physical space than a three-dimensional metamaterial, a cavity or resonator with a metasurface can in principle be made smaller than those that use three-dimensional metamaterials. The phase-matching conditions required at resonance for the separation distance, d , between two metal plates with a metasurface placed in the center is given by [35]

$$d = \frac{\lambda}{2} \left[1 + n\pi - \frac{2}{\pi} \tan^{-1} \left(\frac{\pi}{\lambda} \chi_E \right) \right], \quad (16)$$

$$d = \frac{\lambda}{2} \left[2n\pi - \frac{2}{\pi} \tan^{-1} \left(\frac{\pi}{\lambda} \chi_M \right) \right],$$

for $n = 0, 1, 2, 3, \dots$,

where $n = 0$ is not allowed if $\chi_M \geq 0$ (see [35] for details). From these expressions it is seen that by judiciously choosing the metasurface, it is possible to have a resonator that overcomes the $\lambda/2$ size limit. Various cases for different conditions on the scatterers were given in [35].

As an example, consider a metafilm composed of square patches placed between two metal plates. Figure 13 shows the resonant frequency as a function of l/p (where p is the period and l is the length of the side of one of the squares) for three different plate separations. For reference, the classical result $d = \lambda/2$ occurs when $l/p = 0$. It is seen that the capacitive metasurface can significantly reduce the resonant frequency for a given resonator size d , or equivalently reduce the size of the

resonator needed to obtain a desired resonant frequency. The reduction in resonant frequency of the resonator for the simple square-patch metafilm resonator is shown in Table 1 for $p = 500 \mu\text{m}$. In this table, the percent reduction (which is also a measure of the size reduction possible for a given resonant frequency) is listed for different values of l/p . The results in

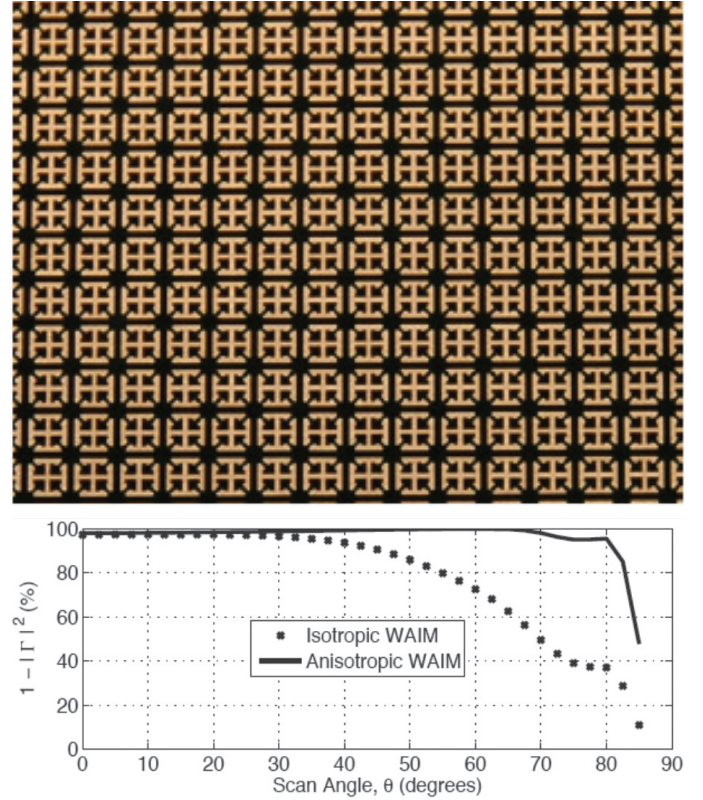


Figure 12. An impedance-matching metasurface: (a) a typical electric-resonator structure; (b) the reflection behavior as a function of angle, after Smith [56].

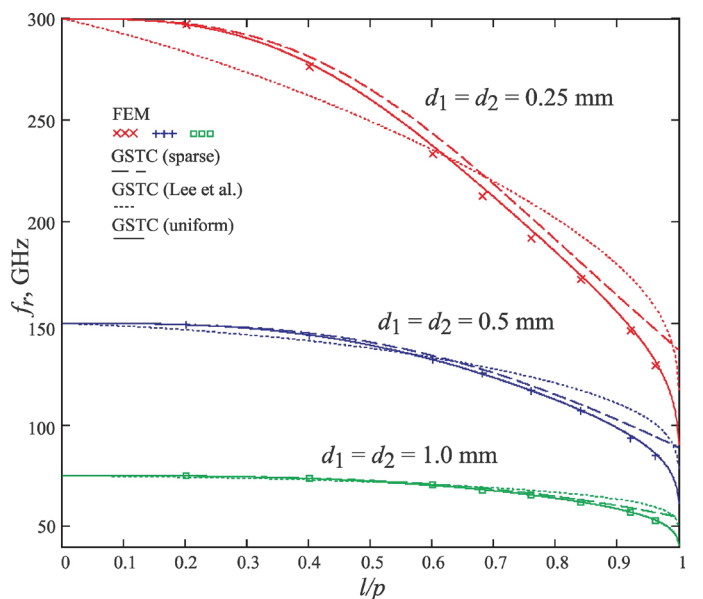


Figure 13. The resonant frequency of an air-filled parallel-plate resonator loaded with a square-patch capacitive metafilm.

Table 1. The reduction in resonant frequency as a function of l/p for a parallel-plate resonator with a metasurface consisting of square metal patches with $p = 500 \mu\text{m}$.

l/p	$d = 0.5 \text{ mm}$	$d = 1.5 \text{ mm}$	$d = 2.0 \text{ mm}$
0	0%	0%	0%
0.4	7.3%	3.8%	1.9%
0.6	20.9%	11.9%	6.4%
0.8	38.4%	25.2%	14.9%
0.96	56.1%	42.6%	29.0%

this table show that a square-patch metafilm placed at the center of a resonator can reduce the size by as much as 56%. If metasurfaces are designed with scatterers having more elaborate polarizability characteristics (e.g., judiciously chosen resonant behavior), it should in principle be possible to achieve even greater reductions in size. In fact, by controlling the properties of the metasurface, a frequency-agile resonator could be realized.

5. Waveguides

Because metasurfaces can be designed to have total reflection of an incident wave, it should be possible to trap and guide electromagnetic energy in a region between two metasurfaces. This class of waveguides is illustrated in Figure 14 (note the change in coordinate system compared to Figure 7, in order to have the waveguide mode propagate along z) [37]. Assuming a waveguide mode with the propagation constant in the z direction, $\beta = k_0 \sin \theta$, from Equation (13) we find that for a TE mode, the normalized propagation constant must obey

$$n_e = \frac{\beta}{k_0} \sqrt{\frac{\chi_{MS}^{zz} \chi_{ES}^{yy} + 4}{- \chi_{MS}^{zz} \chi_{MS}^{xx}}} \quad (17)$$

in order for total reflection to occur. This quantity will generally be complex for leaky modes. Once the scatterers that compose the metasurface are chosen to meet the above criterion and β is determined, the transverse wavenumber in the x direction is given by

$$k_x = \sqrt{k_0^2 - \beta^2} = k_0 \sqrt{1 - n_e^2}, \quad (18)$$

and the separation distance, d , between the two metasurfaces is given by

$$d = \frac{\left(n + \frac{1}{2} \right) \pi + 2 \tan^{-1} \left(\frac{k_x \chi_{MS}^{zz}}{2} \right)}{k_x} \text{ for } n = 1, 2, 3, \dots \quad (19)$$

It is desirable to find conditions for which the imaginary part of n_e is as small as possible, and we need to ensure that $\text{Im}(n_e) < 0$ and $0 < \text{Re}(n_e) < 1$ (the latter conditions because otherwise the mode will basically be a surface wave localized near the two metasurfaces, and will likely suffer increased attenuation as a result). Similar sets of expressions for the TM modes along with an example were given in [37].

This waveguide shows the promise of being compact, with low material and radiation losses. If the metasurfaces are constructed in a polymer type of material, then it should be possible to develop a flexible wave-guiding structure. If the scatterers composing the metafilm are then chosen properly, a flexible low-loss waveguide could be developed, which could have potential applications at terahertz frequencies. It is also possible to control the properties of the scatterers. This could be used to design controllable or smart frequency-agile wave-guiding structures.

6. Complex Modes and Surface Waves on a Metasurface

As with the case of a classical dielectric slab, metasurfaces may also support surface waves under suitable conditions. However, unlike a conventional dielectric slab, forward and backward surface waves, as well as complex modes, can be simultaneously excited on the metafilm, a direct consequence of engineering the properties of the constituent scatterers. In fact, the scatterers that compose a metafilm can be judiciously chosen such that surface waves and/or complex modes will be present only at desired frequencies. Details of when surface waves can be supported on a metasurface were given in [109]. There, it was shown that the poles of the reflection coefficient for a line source (either electric or magnetic) above a metasurface are functions of the surface susceptibilities of the metasurface. Once these poles are determined, the propagation constant for a wave traveling along the surface under various conditions on the electrical and magnetic surface susceptibilities is given by

$$\beta_s = \sqrt{k_0^2 - k_x^2}, \quad (20)$$

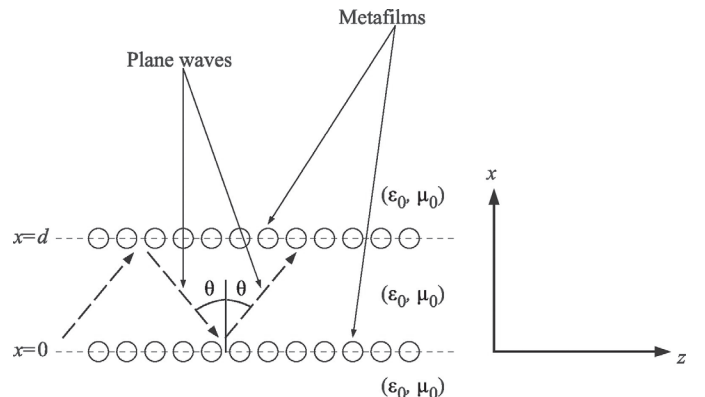


Figure 14. An illustration of a metasurface waveguide.

Table 2. Conditions for surface waves and guided modes, as well as expressions for the propagation constant.

Conditions on χ_{ES}^{yy} , χ_{MS}^{zz} , and χ_{MS}^{xx}		Number of Guided Waves	Guided-Wave Poles	Type of Guided Wave
$\chi_{MS}^{zz} > 0$	$\chi_{MS}^{zz} \geq \frac{1}{k^2 \chi_{MS}^{zz}} - \chi_{MS}^{zz}$	2	$\beta_{1,2} = \frac{-j \pm \sqrt{k^2 \chi_{MS}^{zz} (\chi_{ES}^{yy} + \chi_{MS}^{zz})}}{\chi_{MS}^{zz}}$	complex mode
$\chi_{MS}^{zz} > 0$	$-\chi_{MS}^{zz} < \chi_{ES}^{yy} \leq \frac{1}{k^2 \chi_{MS}^{zz}} - \chi_{MS}^{zz}$	2	$\beta_{1,2} = -j \left[\frac{1 \pm \sqrt{1 - k^2 \chi_{MS}^{zz} (\chi_{ES}^{yy} + \chi_{MS}^{zz})}}{\chi_{MS}^{zz}} \right]$	surface wave
$\chi_{MS}^{zz} > 0$	$\chi_{ES}^{yy} \leq -\chi_{MS}^{zz}$	1	$\beta_1 = -j \left[\frac{1 + \sqrt{1 - k^2 \chi_{MS}^{zz} (\chi_{ES}^{yy} + \chi_{MS}^{zz})}}{\chi_{MS}^{zz}} \right]$	surface wave
$\chi_{MS}^{zz} > 0$	$\chi_{ES}^{yy} = 0$	2	$\beta_{1,2} = -j \left[\frac{1 \pm \sqrt{1 - k^2 (\chi_{MS}^{zz})^2}}{\chi_{MS}^{zz}} \right]$	surface wave
$\chi_{MS}^{zz} < 0$	$\chi_{ES}^{yy} < -\chi_{MS}^{zz}$	1	$\beta_1 = -j \left[\frac{1 - \sqrt{1 - k^2 \chi_{MS}^{zz} (\chi_{ES}^{yy} + \chi_{MS}^{zz})}}{\chi_{MS}^{zz}} \right]$	surface wave
$\chi_{MS}^{zz} = 0$	$\chi_{ES}^{yy} > 0$	1	$\beta_1 = -\frac{jk^2}{2} \chi_{ES}^{yy}$	surface wave

where β_s is given in Table 2 for various conditions for an electric line source; values for a magnetic line source were given in [109]. When the surface susceptibilities of the metasurface meets one of these conditions, surface waves or complex modes can be excited. Figure 15 illustrates a metasurface that has either a surface wave or complex modes, depending on the frequency of excitation. This figure shows the magnitude of the $\mathbf{E}$ field for a magnetic line source placed above an array of spherical particles (see [109] for the geometry). Figure 15a illustrates the presence of one surface wave, and Figure 15b illustrates the case for a complex mode. By varying the properties of the scatterers, the surface susceptibilities will change. Hence, from the results in Table 2, it is in principle possible to have a surface wave and/or complex mode exist at any desired frequency, i.e., to realize a frequency-agile guided-wave structure.

7. Terahertz Devices

Controllable surfaces may prove to be beneficial for the development of components and devices at terahertz frequencies. In essence, controllable metasurfaces are created by

altering the electrical landscape, configuration, or environment of the elements composing the surface. When the electrical landscape is changed – usually by means of a semiconductor inclusion in a scatterer – these are sometimes termed hybrid metamaterials [38]. The terahertz (THz) frequency regime is one area in which controllable metasurfaces have made a big impact. This is due mainly to the dearth of practical THz technology. Devices considered commonplace in microwave and photonic areas, such as switches and modulators, largely don't exist in the THz regime. A major challenge has been to find natural materials that respond strongly and controllably to THz radiation, yet do not suffer high loss.

Numerous demonstrations have now shown that THz metasurfaces offer these advantages. THz metamaterials leverage the benefits of a high ability to be tuned, relatively easy fabrication, low losses, and dynamic control. The first dynamic control over a THz metasurface utilized a near-infrared laser that actively switched off the resonant response by increasing the metasurface's substrate conductivity [39]. The concept utilized copper split-ring resonators (SRRs), fabricated on high-resistivity GaAs. When illuminated by amplified laser light, conduction electrons were excited across the bandgap of the GaAs, making it semi-metallic. This shorted out the split-ring resonator's capacitive gap, turning off the

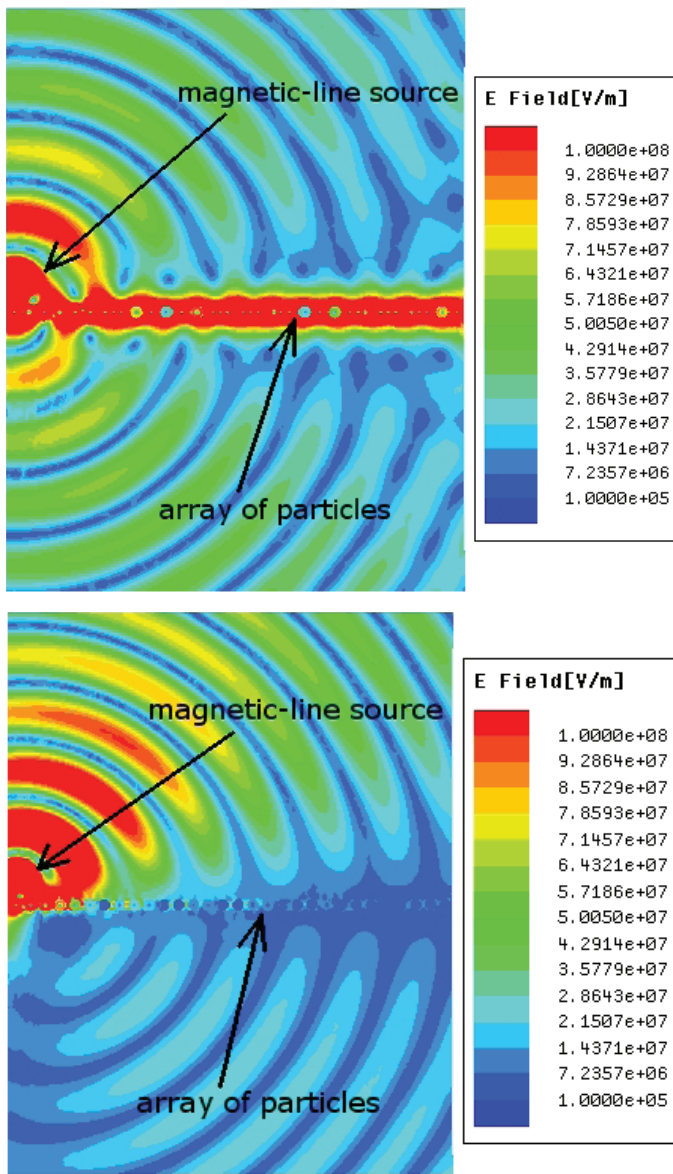


Figure 15. The magnitude of the E field (on a linear scale) from a magnetic line source placed 45.49 mm above an array of spherical particles: (a) $f = 1.42$ GHz, one surface wave; and (b) $f = 1.5$ GHz, complex mode.

fundamental resonance and changing the metasurface's macroscopic response. The concept was also demonstrated with short-carrier-lifetime semiconductors, enabling ultra-fast modulation [40]. These types of “switches” have demonstrated switching speeds of 20 ps. Optical control was soon supplemented by electronic control schemes. Here, an array of gold split-ring resonators formed a Schottky junction with a moderately doped GaAs layer grown by molecular-beam epitaxy (MBE). The entire metamaterial array was surrounded by an ohmic contact, providing the second electrical connection. In the unperturbed state, the metasurface had no resonance, because the resonators were shorted by the doped GaAs. Upon application of a reverse voltage bias, the split-ring resonators formed an increasingly large depletion region in the GaAs near their gap. This formed an insulating region around the split-ring resonators, reestablishing their resonant response, and thus changing the macroscopic properties [41]. It was realized later

that amplitude modulation in this method is always accompanied by phase modulation at different frequencies, according to the Kramers-Kronig relations. As such, even though the split-ring resonator resonance is narrowband, a modulation effect (in one form or another) exists across a wide band. This led to the first demonstration of a solid-state THz chopper, the modulation frequency of which can be as high as 30 kHz [42], a significant improvement over mechanical devices that are limited to ~ 1 kHz for THz beams. These examples vividly illustrated dramatic improvements over existing technologies in switching and modulation of freely propagating THz waves at room temperature.

Many other THz modulators have been developed on the controllable metasurface paradigm. These include a frequency-agile metasurface, in which the resonance frequency shifts upon stimulation by near-infrared light [43]. Here, a small region of silicon is fabricated into the gap of a split-ring-resonator structure (see Figure 16). In its unperturbed state, the silicon

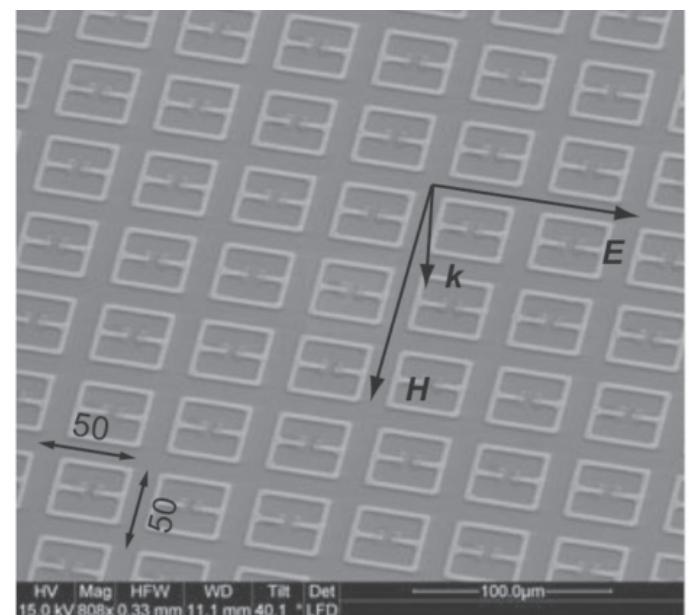
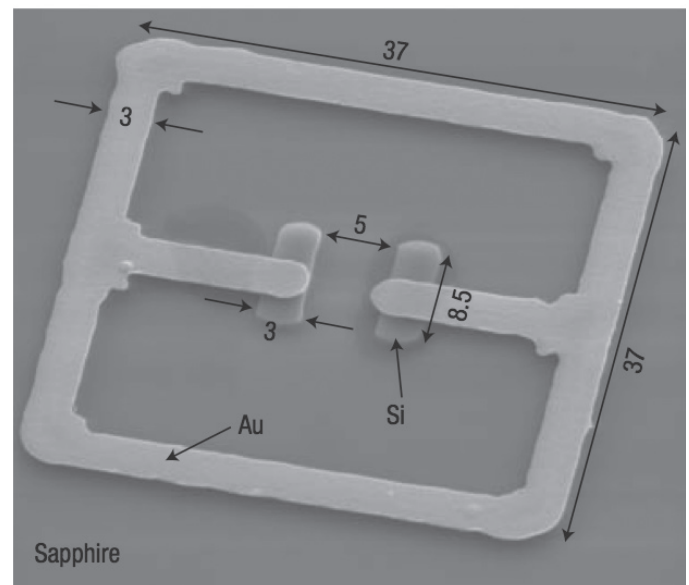


Figure 16. A split-ring resonator with silicon on the gap arms.

is insulating, and thereby contributes only a small amount of capacitance to the overall resonance. Upon optical stimulation, the region becomes semi-metallic and effectively increases the size of the split-ring-resonator “capacitor,” thus lowering the resonance frequency. The novelty here was that one need not destroy the resonance to alter it, as in the previous case. Instead, the resonance can be tuned. Not only did this demonstrate a dynamically tunable resonance, but it also enabled a new form of frequency-shift keying, wherein broadband incident THz radiation is controllably converted to a narrowband output occupying one of two narrow frequency bins.

Tunable THz metasurfaces continue to be developed for new applications and using new physical mechanisms. Amplitude modulators have been adapted to more-specific applications, such as spatial light modulators [44] and quantum-cascade laser modulators [45]. Tunable metamaterials have also been created by other methods, such as the MEMS [46], wherein split-ring resonators are fabricated on bi-material cantilevers that reorient the split-ring resonator with respect to the incident wave upon a change in temperature. Other thermal tuning methods have been investigated. In one, temperature variations changed the intrinsic carrier density of a semiconductor, thus altering its relative permittivity in the THz range. Metasurfaces fabricated on such a substrate would be frequency-agile as a function of temperature [47]. In another variation, vanadium dioxide served as the metasurface substrate [48]. Upon temperature variations, vanadium dioxide undergoes a metal-insulator transition, dramatically changing its dielectric properties, and thus the macroscopic metamaterial resonance. A very interesting development called the memory metamaterial [49] was developed from this work. Here, the hysteretic nature of the vanadium dioxide metal-insulator transition enabled a persistence in the dielectric properties over a certain temperature range. The THz metasurface, being fabricated on this substrate, acquired this same persistent behavior in terms of its resonance frequency, thus allowing a form of memory in the electromagnetic response.

In still other work, THz metasurfaces have been dynamically tuned on a much slower timescale by changing the environment around the resonators through the application of a dielectric coating to the metasurface’s surface [50-52]. This was investigated as a form of remote sensing, based on the idea that the metasurface resonance is very sensitive to changes in the dielectric environment, particularly around the split-ring resonator’s gap. This idea has been extending microwave frequencies where fluids have been used for controllable surfaces, bio-sensing, and microwave-assisted chemistry. This is discussed in detail below.

8. Optical Metasurfaces

Up to this point, we have discussed only RF, microwave, and THz applications of metasurfaces. A few comments are in order on the use of optical-frequency metasurfaces. As at RF and microwave frequencies, optical metamaterials have gained much attention over recent years. The realization

of freely tailoring the electric and magnetic response of a material at optical frequencies holds promise for new and exotic optical phenomena, including optical magnetism [68], a negative index of refraction [4, 69, 70], and ideal lensing [70]. At optical frequencies, plasmonic resonances, typically excited in gold and silver nanostructures, provide a means for controlling the electric and magnetic moments that collectively are responsible for the behavior of a metamaterial. Examples of such passive structures include coupled plasmonic nanorods [69, 70] and spheres [72], voids in metal films [69], metallic fishnet structures [73], and double and single split-ring resonators [68, 71]. Due to the inherent high loss associated with plasmonic materials at optical frequencies, optical metamaterials are severely limited in their practical application. Structures that incorporate gain media to overcome plasmonic losses have therefore also been investigated. Examples include active plasmonic-coated nanoparticles [75], passive plasmonic particles embedded in gain media [69], and gain-coated plasmonic nanoparticles [76]. Such active structures show the potential for new and interesting properties and devices, such as optically-switchable frequency-selective surfaces [77], and the surface-plasmon amplification by stimulated emission of radiation (SPASER) [76]. Another investigating area is the use of optical metasurfaces and metamaterials for so-called nano-transmission lines [78, 79].

Due to fabrication obstacles, optical metamaterials are often limited in construction to single-layer two-dimensional arrays of scatterers [69, 70, 80], i.e., metasurfaces. Much of the published work on optical metamaterials thus actually describes optical metasurfaces. As discussed above, defining bulk properties for these optical metasurfaces must be approached with caution, and generalized sheet-transition conditions provide a preferred means for uniquely describing a two-dimensional array of scatters used for a large range of optical applications. They are therefore a more appropriate approach for describing optical metasurfaces, rather than attempting to define and assign properties more appropriate for describing a bulk material.

Even though most so-called optical metamaterials to date have in fact been metasurfaces, we should mention that recent advances have been made towards realizing truly three-dimensional optical metamaterials [80, 81], where bulk quantities such as the permittivity, permeability, and index of refraction can be appropriately and uniquely defined. Improved fabrication techniques have made possible the layering and stacking of metasurfaces into volume arrays of scatterers. For example, negative refraction of light was recently observed in an optical prism made of negative-index material [81].

9. Microfluidics for Controllable Surfaces, Assisted Chemistry, and Biomedical Sensors

One of the disadvantages often mentioned about metamaterials (and metasurfaces) is that they have a very narrow bandwidth of operation (less than a few percent) in the

desired frequency range. While true, rather than being a limitation, this narrowband behavior can be utilized to advantage in some applications. Three such applications are in the areas of fluid-controllable surfaces, microwave-assisted chemistry, and biomedical sensors.

9.1 Fluid-Controllable Surfaces

The highly resonant nature of metamaterials and metasurfaces provides a means for tuning the frequency response of these structures. Perturbations to the electric or magnetic response of the inclusions making up a metasurface can be modified in real time, thereby altering the effective material response. This was discussed above in Section 2, where a magnetic-bias field was used to change the polarizabilities of magneto-dielectric spheres in a metasurface, and in Section 7 for THz applications. However, many metamaterials and metasurfaces are constructed from metallic inclusions that have plasma resonances governed largely by the choice of their geometry. Aside from the specific geometry, the polarizabilities of these metallic inclusions may also be controlled by affecting the capacitive and or inductive properties that dictate the plasma resonances. The electric-coupled resonator is a structure that provides a means for directly perturbing the capacitive response by altering the electrical properties of the material occupying the electrical gap. One way to accomplish this is to cause different fluids to flow in a channel over the gap: for details, see [82].

Figure 17 shows a diagram of the type of electric-field-coupled-resonator inclusion (referred to as an OE2 structure [29]) that was used in designing a fluid-controllable metasurface for operation in the S band over 2.6 GHz to 3.9 GHz, with the following dimensions: $t = w = 0.5$ mm, $d = 9.5$ mm, $l = 5$ mm, and $g = 0.15$ mm. This type of inclusion is easily excited by the electric field of the incident wave, if the inclusions are suitably oriented.

Being able to orient the inclusions in the plane of the metasurface has several attractive features: (1) it is consistent with the planar fabrication techniques used for microwave circuits and polydimethylsiloxane (PDMS) microfluid channels; (2) it allows the capacitive gaps for each inclusion to be simultaneously aligned with each other as well as with the fluid channel, therefore allowing multiple inclusions to share the same fluid-channel path; and (3) it allows the fluid channels to be in direct contact with the capacitive gaps of the inclusions, which facilitates strong coupling of the fluid channels to the electric field that is excited in the gaps.

A diagram of a fluid-tunable metasurface made of a 3×6 square array of inclusions is shown in Figure 18. The period of the inclusions was 11 mm. The simulated response of this metasurface placed in a 72 x 34 mm S-band waveguide is shown in Figure 19. The simulations discussed here were performed by use of Ansoft *HFSS* (mention of this product is not an endorsement, but serves only to identify the software used).

Experiments on a 3×6 array of these inclusions, fabricated from Duroid 6002 High-Frequency Laminate (mention of this product is not an endorsement, but serves only to specify the material used) with 17 μ m-thick copper laminated to a 0.508 mm-thick substrate, was used to investigate fluid tuning. Polymer tubing was used for fluid channels. The polymer fluid channels were bonded to the metasurface by use of a cyanoacrylate-based glue. A picture of the metasurface with the

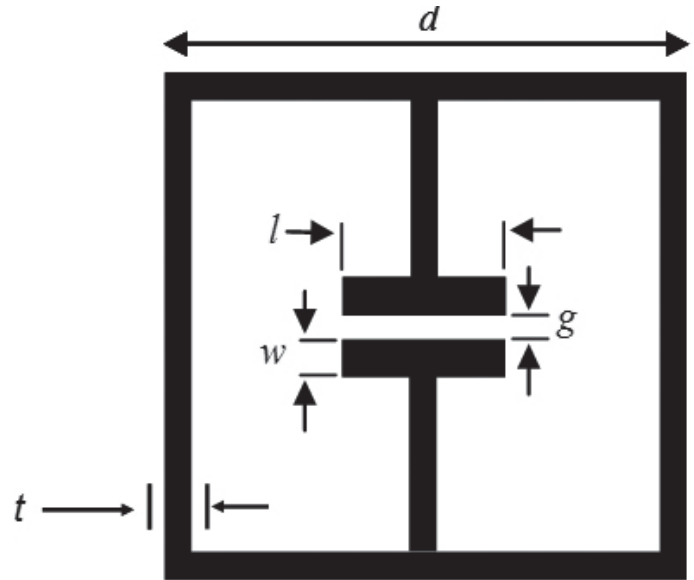


Figure 17. An electric-coupled resonator (OE2 structure).

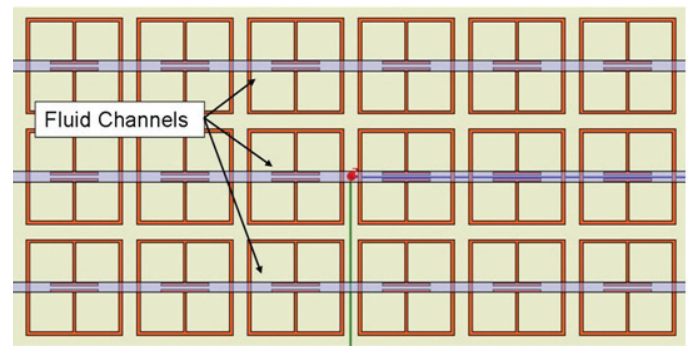


Figure 18. A fluid-tunable metafilm, showing the metafilm array and fluid-channel orientation.

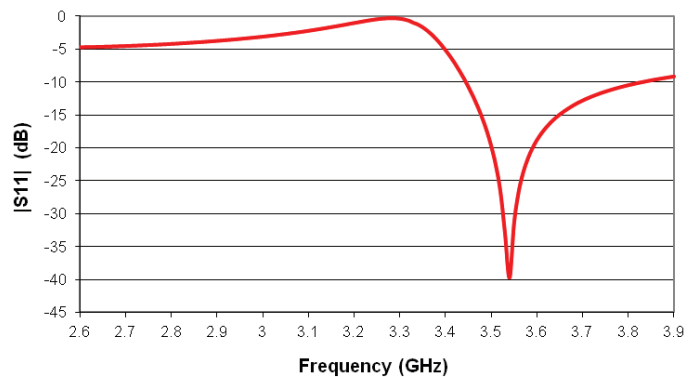


Figure 19. The simulated response of a metafilm composed of a 3×6 array of OE2 inclusions in an S-band waveguide.

polymer fluid channels bonded and aligned across the gaps in the inclusions is shown in Figure 20a. Figure 20b illustrates a second structure fabricated from gold on glass with PDMS (polydimethylsiloxane) fluid-channel sections.

The 3×6 metasurface array was supported by a block of polystyrene foam that filled two-thirds of the waveguide cross section, and was used as a handle so that the metasurface could be easily taken in and out, in order to facilitate quick filling of the fluid channels. The input and output ports of the waveguide were then connected to a calibrated vector network analyzer. Employing the TRL (thru-reflect-line) calibration method [110], we measured the scattering parameters of the waveguide over the frequency range of 2.6 GHz to 3.95 GHz. The measurement uncertainties in reflection coefficient resulting from this calibration method were $\Delta|S_{11}| = 0.02$ (one standard deviation).

Measurements with and without fluid in the channels were performed to determine the tuning of the transmission resonance. De-ionized water with a relative permittivity of $\epsilon_r = 81$ was chosen to test the ability to tune the frequency of the metasurface, due to its ease of handling, low volatility, high dielectric constant, and low loss in the S band. Fluid channels were manually filled by use of a syringe. Figure 21 shows the metasurface inside the S-band waveguide during the experiment.

Experimental results are shown in Figure 22. They demonstrated the ability to tune over a frequency range of 150 MHz. The reflection dip was clearly shifted from 3.75 GHz down to 3.6 GHz between the cases without and with water in the fluid channels, respectively. The disagreement between the positions of the resonance dip for the case without water and that of the simulated result in Figure 19 is believed to be due to inclusion-fabrication errors. The additional oscillations in the response are also a result of fabrication nonuniformity and periodicity errors of the inclusions, as well nonuniform perturbations across the metasurface due to errors in bonding the fluid channels. In the future, fabrication improvements will be explored to mitigate these errors.

In addition to using fluids for specifically tuning microwave characteristics as focused on here, changes in resonance characteristics may also be used as a means to monitor processes tied to the changes in the fluid flowing in the channels. This approach has applications to manufacturing, industrial, biomedical, and chemical processes. The use of metamaterials/metasurfaces for influencing and sensing external process is discussed in greater detail below.

9.2 Microwave-Assisted Chemistry Applications

A metafilm operating in the resonant state stores electromagnetic energy in the modes of the inclusions. This property can be exploited to increase the interactions experienced between the electromagnetic fields and liquids carried by the

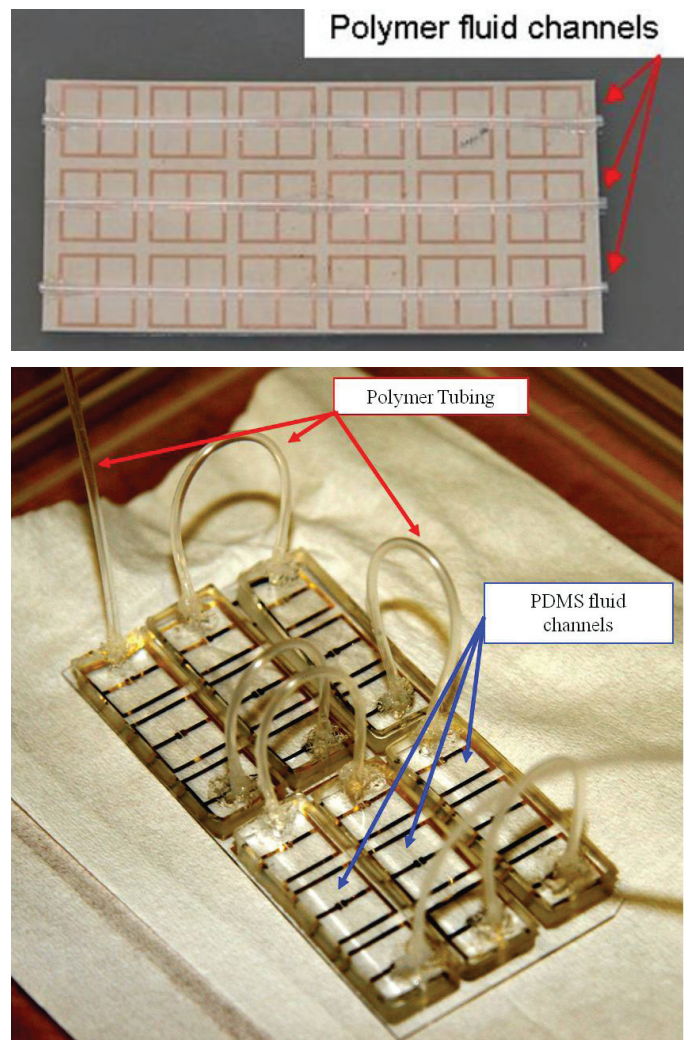


Figure 20. A fluid-tunable metafilm composed of a 3×6 array of OE2 resonators: (a) fabricated from copper on Duroid with three polymer-tubing fluid-channel sections; (b) fabricated from gold on glass with PDMS fluid-channel sections.

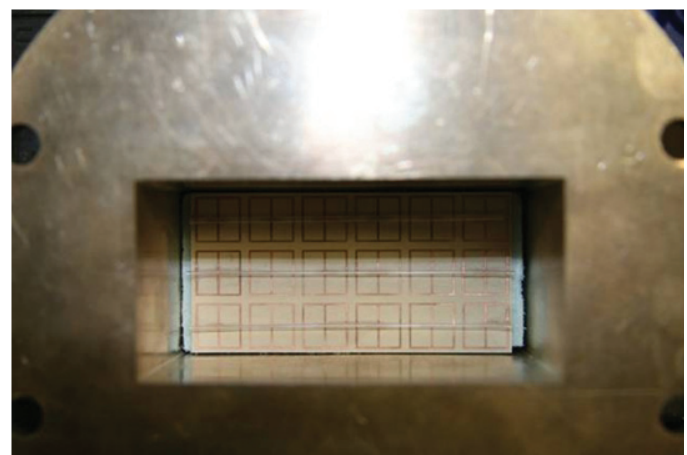


Figure 21. A 3×6 array of OE2 resonators fabricated from copper on Duroid with three polymer-tubing fluid-channel sections placed in an S-band waveguide.

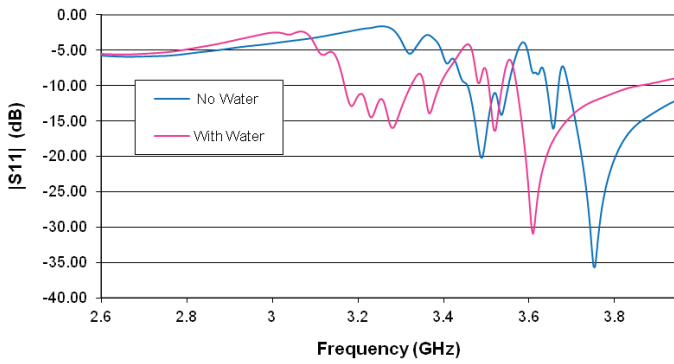


Figure 22. Experimental results comparing S_{11} for the copper on Duroid microfluidic metafilm for empty fluid channels, and for fluid channels filled with de-ionized water over the S band (2.6-3.95 GHz).

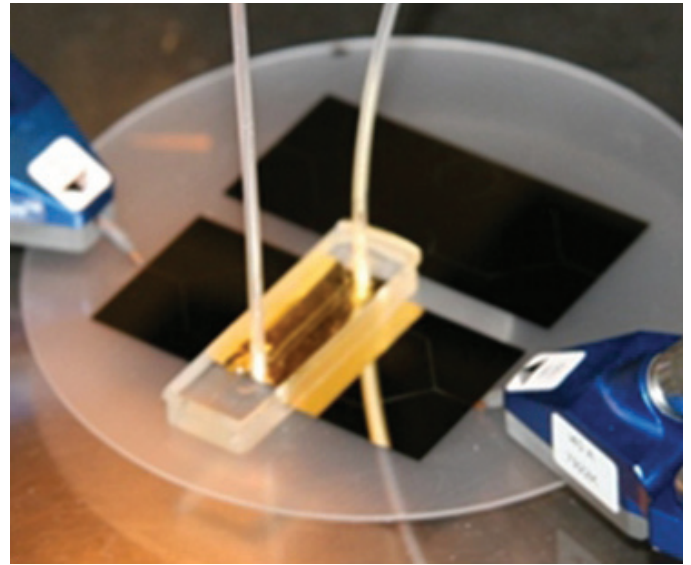
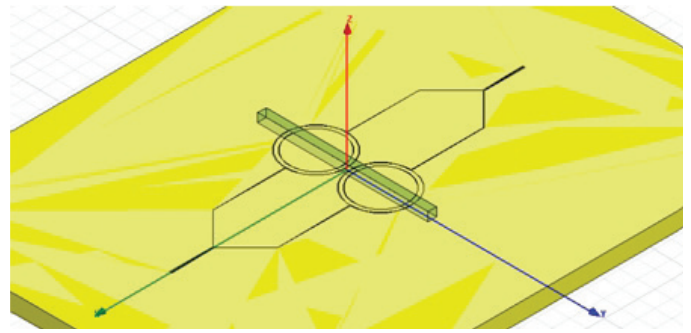


Figure 24. A split-ring-resonator coplanar-waveguide bio-sensor concept.

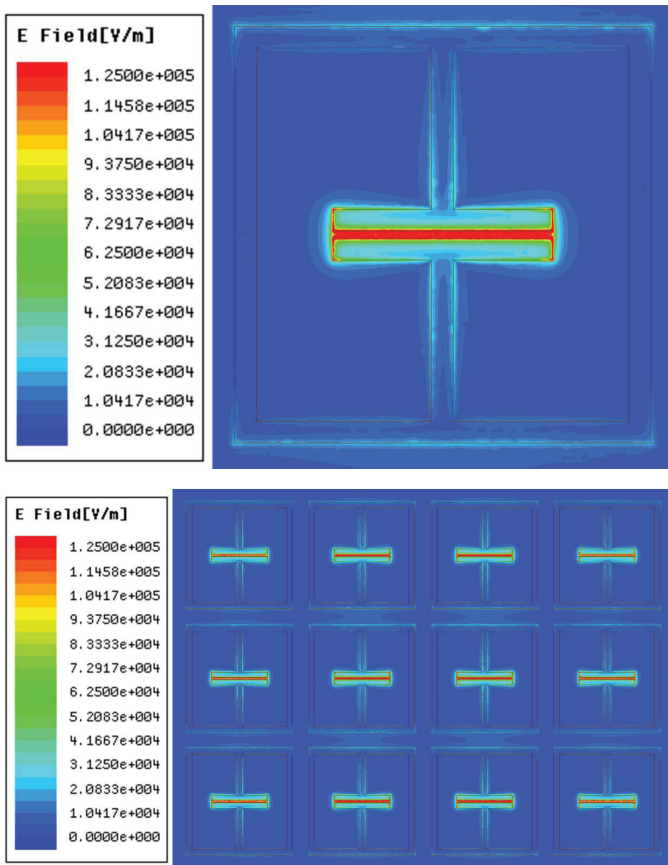


Figure 23. The electric-field structure in the gap of the inclusions of the metasurface showing the enhancement of the electric field. The maximum field amplitude was 1.25×10^5 V/m at 3.29 GHz.

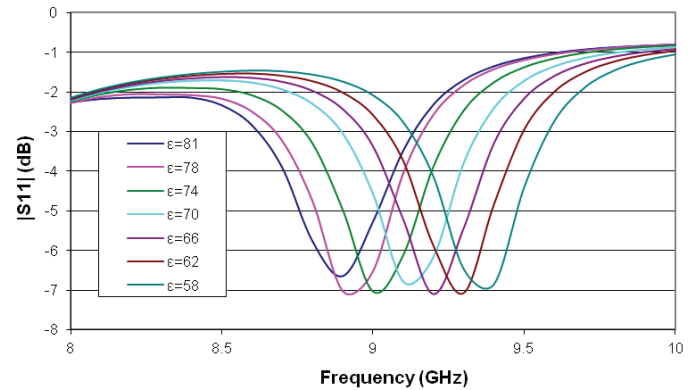


Figure 25. A simulated response, showing S_{11} for the split-ring-resonator coplanar-waveguide bio sensor as the relative permittivity of the fluid in the channel was reduced from 81 to 58.

fluidic channels. There is currently an explosion of research into processes that use microwave energy to catalyze chemical or biochemical reactions [111-113], and many of these types of processes could benefit from the use of metafilms to increase the coupling between electromagnetic fields and fluids. The metafilm approach described here is particularly valuable for reactions that are implemented by use of a continuous-flow geometry, as chemical composition can be precisely controlled by controlling reactant flow rates, while the amount of energy transferred to the fluid can be precisely controlled by adjusting the frequency and power of the microwave excitation, in addition to controlling the flow rate of the fluid across the metafilm. In addition, because the resonant frequency of the metafilm array can be tailored by the geometry and arrangement of the inclusions, different metafilms with different resonant frequencies can be stimulated by a single variable-frequency source within a single waveguide. The metasurface structure used above to demonstrate fluid tuning was also investigated as a means for concentrating electric fields in a waveguide [82].

Simulations of the electric-field strength inside the waveguide when the fundamental mode of the S-band waveguide was excited were conducted. For a 1 W input at 3.29 GHz, the maximum field strength calculated was 8×10^2 V/m. For the same waveguide, but with the metasurface placed in the center, the maximum electric-field strength calculated in the gaps of the electric-field coupled inclusions was 1.25×10^5 V/m. This demonstrated a field enhancement of at least two orders of magnitude, and an enhancement in the specific absorption rate (SAR) of at least four orders of magnitude. The field structure in the central gap of the inclusion is shown in Figure 23.

The localized field structure shown in Figure 23 demonstrated the possibility of using a metamaterial structure to precisely deliver electromagnetic energy. The fluid-tunable metasurface discussed above clearly demonstrated the interaction of a fluid channel with the field in the gaps of the inclusions comprising the metasurface. It can therefore be envisioned that such a metasurface coupled to a fluidic network could be used to precisely deliver microwave fields for assisting in chemical processes confined within the fluid channels.

9.3 Biosensor Applications

The concept of the fluid-tunable metasurface discussed above can be extended to realize highly resonant integrated and chip-level structures for sensor applications. Integrating single inclusions used in realizing a metamaterial into a circuit or network may allow for compact sensors for use in biomedical applications to count cells or monitor reactions in fluids. An example of such a device is shown in Figure 24. Two splitting resonators are coupled to a coplanar-waveguide (CPW) structure. A fluid channel is coupled to the gaps in the splitting resonators in much the same way as the fluid channels of the fluid-tunable metasurface above were used to affect the capacitance of the inclusions. By altering the electrical prop-

erties of the fluid in the channels, the transmission characteristics of the coplanar-waveguide structure can be tuned.

Simulation results for this structure are shown in Figure 25, where the relative permittivity of the fluid in the fluid channel was changed from $\epsilon_r = 81$ (that of water) to lower permittivities, down to $\epsilon_r = 58$. As the fluid was changed, the reflection coefficient was shifted in frequency over several tens of MHz. This ability to affect the resonance characteristics of a metamaterial-inspired structure using a fluid channel provides a means for realizing a compact sensor, suitable for biomedical-sensing applications such as counting cells or particles in a fluid. Future work includes experimental verification of this type of sensor, as well as determining the minimum detectable change in permittivity, and the feasibility of counting particles with such a device. Other types of compact resonator structures will be explored as well, which may be suitable for integrated-sensor devices. We are currently investigating the possibility of using this concept for detecting and counting small particles in fluids, e.g., counting white blood cells.

10. Conclusion

Metasurfaces offer an alternative to bulk three-dimensional metamaterials. Because of the two-dimensional nature of the metasurface structures, they occupy less physical space and can exhibit lower loss. We have pointed out that while effective-property models are valid for three-dimensional metamaterials, the more appropriate way to characterize metasurfaces/metafilms is by their effective surface electrical and magnetic susceptibilities, where these surface susceptibilities are related to the scatterers that compose the metasurface. In this overview paper, we have discussed various applications of metasurfaces, ranging from microwave to optical frequencies. The applications discussed here are by no means the only applications possible. However, they offer a few ideas of where metasurfaces can be used, and hopefully the paper opens the door to the development of other new applications. The analysis tools presented here give one the means to model, conceive, and generate new applications in the future.

What about a one-dimensional metamaterial? The two-dimensional metamaterial (i.e., a metasurface) concept can be extended even further to the concept of using only a linear unit cell, rather than a square, a rectangle, etc., that is, using only a single sub-wavelength resonant structure for some desired effect or behavior. In fact, we have already begun to see a few applications of this concept. One in particular is the use of a unit cell in the design of electrically small antennas: for details, see [114-121]. In the antenna application, the unit-cell acts like a parasitic element to the radiating element of the antenna, and serves as a means to match the electrically small radiating element to both (1) the feeding transmission line and (2) free space. Such designs have been shown to achieve efficient electrically small antennas [114-121]. Nanoparticles have also been used for tuning so-called optical nanoantennas [122]. Another example is using a one-dimensional unit cell as

a tuning structure for planar transmission lines [123], as well as the sensor shown in Figure 24. Another emerging area of application is the use of one-dimensional chains of nanoparticles as waveguides supporting surface waves, of which examples could be found in [124-130].

While much more work is still needed in the understanding, analysis, design, and fabrication of both metamaterials and metasurfaces, we see that the development of such materials and surfaces in recent years is bringing us closer to realizing the exciting predictions made over one hundred years ago by the work of Lamb, Schuster, and Pocklington [18-20]. The potential of these materials has changed forever the landscape of RF, microwaves, optics, and photonics for the future.

11. References

1. S. Zouhdi, A. Sihvola and M. Arsalane (eds.), *Advances in Electromagnetics of Complex Media and Metamaterials*, Boston, Kluwer Academic Publishers, 2002.
2. N. Engheta and R. W. Ziolkowski, *Electromagnetic Metamaterials: Physics and Engineering Explorations*, Hoboken, NJ, John Wiley & Sons, 2006.
3. G. V. Eleftheriades and K. G. Balmain, *Negative Refraction Metamaterials: Fundamental Principles and Applications*, Hoboken, NJ, John Wiley & Sons, 2005.
4. V. G. Veselago, "The Electrodynamics of Substances with Simultaneously Negative Values of ϵ and μ " [in Russian], *Usp. Fiz. Nauk*, **92**, 1967, pp. 517-526; English translation in *Sov. Phys. Uspekhi*, **10**, 1968, pp. 509-514.
5. D. R. Smith, W. J. Padilla, D. C. Vier, S. C. Nemat-Nasser and S. Schultz, "Composite Medium with Simultaneously Negative Permeability and Permittivity," *Phys. Rev. Lett.*, **84**, 2000, pp. 4184-4186.
6. R. Marques, J. Martel, F. Mesa and F. Medina, "A New 2D Isotropic Left-Handed Metamaterial Design: Theory and Experiment," *Micr. Opt. Technol. Lett.*, **35**, 5, 2002, pp. 405-408.
7. C. L. Holloway, E. F. Kuester, J. Baker-Jarvis and P. Kabos, "A Double Negative (DNG) Composite Medium Composed of Magneto-Dielectric Spherical Particles Embedded in a Matrix," *IEEE Transactions on Antennas and Propagation*, **AP-51**, 10, 2003, pp. 2596-2603.
8. A. Sihvola, "Metamaterials in Electromagnetics," *Metamaterials*, **1**, 1, 2007, pp. 2-11.
9. E. Shamonina and L. Solymar, "Metamaterials: How the Subject Started," *Metamaterials*, **1**, 1, 2007, pp. 12-18.
10. M. Lapine and S. Tretyakov, "Contemporary Notes on Metamaterials," *IET Microwaves, Antennas, and Propagation*, **1**, 1, 2007, pp. 3-11.
11. D. V. Sivukhin, "The Energy of Electromagnetic Fields in Dispersive Media" [in Russian], *Opt. Spektrosk.*, **3**, 1957, pp. 308-312.
12. G. D. Malyuzhinets, "A Note on the Radiation Principle" [in Russian], *Zh. Tekh. Fiz.*, **21**, 1951, pp. 940-942.
13. R. A. Silin, "Optical Properties of Artificial Dielectrics (Review)" [in Russian], *Izv. VUZ Radiofiz.*, **15**, 1972, pp. 809-820; English translation in *Radiophys. Quantum Electron.*, **15**, 1972, pp. 615-624.
14. R. A. Silin, "Possibility of Creating Plane-Parallel Lenses," [in Russian], *Opt. Spektrosk.*, **44**, 1978, pp. 189-191; English translation in *Opt. Spectrosc.*, **44**, 1978, pp. 109-110.
15. R. A. Silin and I. P. Chepurnykh, "On Media with Negative Dispersion" [in Russian], *Radiotekh. Elektron.*, **46**, 2001, pp. 1212-1217; English translation in *J. Commun. Technol. Electron.*, **46**, 2001, pp. 1121-1125.
16. L. I. Mandel'shtam, "Lectures on Certain Problems of Oscillation Theory: Lecture 4" [in Russian], *Polnoe Sobraniye Trudov*, **5**, Leningrad, Izdat. Akad. Nauk SSSR, 1950, pp. 461-467; also in his *Lektsii po Optike Teorii Otnositel'nosti i Kvantovoi Mekhanike*, Moscow, Nauka, 1972, pp. 431-437.
17. L. I. Mandel'shtam, "Group velocity in crystalline arrays" [in Russian], *Zh. Eksp. Teor. Fiz.*, **15**, 1945, pp. 475-478; also in *Polnoe Sobraniye Trudov*, **2**, Leningrad, Izdat. Akad. Nauk SSSR, 1947, pp. 334-338.
18. H. Lamb, "On Group-Velocity," *Proc. London Math. Soc.*, Ser. 2, **1**, 1904, pp. 473-479.
19. A. Schuster, *An Introduction to the Theory of Optics*. London, Edward Arnold, 1904, pp. 313-318.
20. H. C. Pocklington, "Growth of a Wave-Group when the Group Velocity is Negative," *Nature*, **71**, 1905, pp. 607-608.
21. L. Lewin, "The Electrical Constants of a Material Loaded with Spherical Particles," *J. IEE (London)*, III, **94**, 1947, pp. 65-68.
22. R. Gans and H. Happel, "Zur Optik kolloidaler Metallösungen," *Ann. Physik*, **29**, 7, 1909, pp. 277-300.
23. E. F. Kuester, M. A. Mohamed, M. Piket-May and C. L. Holloway, "Averaged Transition Conditions for Electromagnetic Fields at a Metafilm," *IEEE Transactions on Antennas and Propagation*, **AP-51**, 10, 2003, pp. 2641-2651.

24. C. L. Holloway, M. A. Mohamed, E. F. Kuester and A. Dienstfrey, "Reflection and Transmission Properties of a Metafilm: With an Application to a Controllable Surface Composed of Resonant Particles," *IEEE Trans. Electromag. Compat.*, **47**, 4, 2005, pp. 853-865.
25. C. L. Holloway, P. Kabos, M. A. Mohamed, E. F. Kuester, J. Gordon, M. D. Janezic and J. Baker-Jarvis, "Realization of a Controllable Metafilm/Metasurface Composed of Resonant Magnetodielectric Particles: Measurements and Theory," *IET Microwaves, Antennas, and Propagation*, **4**, 8, 2010, pp.1111-1122.
26. J. F. O'Hara, E. Smirnova, A. K. Azad, H.-T. Chen and A. J. Taylor, "Effects of Microstructure Variations on Macroscopic Terahertz Metafilm Properties," *Active and Passive Electronic Components*, **2007**, 2007, art. 49691.
27. W. J. Padilla, D. N. Basov and D. R. Smith, "Negative Refractive Index Metamaterials," *Materials Today*, **9**, 7, 2006, pp. 28-35.
28. H. O. Moser, B. D. F. Casse, O. Wilhelmi and B. T. Saw, "Terahertz Response of a Microfabricated Rod-Split-Ring-Resonator Electromagnetic Metamaterial," *Phys. Rev. Lett.*, **94**, 2005, art. 063901.
29. W. J. Padilla, M. T. Aronsson, C. Highstrete, M. Lee, A. J. Taylor and R. D. Averitt, "Electrically Resonant Terahertz Materials: Theoretical and Experimental Investigations," *Phys. Review B*, **75**, 2007, 041102.
30. D. R. Smith, D. Schurig and J. J. Mock, "Characterization of a Planar Artificial Magnetic Metamaterial Surface," *Phys. Review E*, **74**, 2006, art. 036604.
31. C. Highstrete, M. Lee and W. Padilla, "Complementary Planar Terahertz Metamaterials," *Optics Express*, **15**, 3, 2007, pp. 1084-1095.
32. Z. Q. Li, G. M. Wang, K. J. Mikolaitis, D. Moses, A. J. Heeger and D. N. Basov, "An Infrared Probe of Tunable Dielectrics in Metal-Oxide-Semiconductor Structures," *Appl. Phys. Lett.*, **86**, 2005, art. 223506.
33. W. J. Padilla, A. J. Taylor, C. Highstrete, M. Lee and R. D. Averitt, "Dynamical Electric and Magnetic Metamaterial Response at Terahertz Frequencies," *Phys. Review Lett.*, **96**, 2006, art. 107401.
34. H.-T. Chen, W. J. Padilla, J. M. O. Zide, S. R. Bank, A. C. Gossard, A. J. Taylor and R. D. Averitt, "Ultrafast Optical Switching of Terahertz Metamaterials Fabricated on ErAs/GaAs Nanoisland Superlattices," *Optics Lett.*, **32**, 12, 2007, pp. 1620-1622.
35. C. L. Holloway, D. C. Love, E. F. Kuester, A. Salandrino and N. Engheta, "Sub-Wavelength Resonators: On the Use of Metafilms to Overcome the $\lambda/2$ Size Limit," *IET Microwaves, Antennas, and Propagation*, **2**, 2, 2008, pp. 120-129.
36. M. Caiazzo, S. Maci, and N. Engheta, "A Metamaterial Surface for Compact Cavity Resonators," *IEEE Antennas and Wireless Propagation Letters*, **3**, pp. 261-264, 2004.
37. C. L. Holloway, E. F. Kuester and D. Novotny, "Waveguides Composed of Metafilms: The Two-Dimensional Equivalent of Metamaterials," *IEEE Antenna and Wireless Propagation Letters*, **8**, 2009, pp. 525-529.
38. H.-T. Chen, S. Palit, T. Tyler, C. M. Bingham, J. M. O. Zide, J. F. O'Hara, D. R. Smith, A. C. Gossard, R. D. Averitt, W. J. Padilla, N. M. Jokerst and A. J. Taylor, "Hybrid Metamaterials Enable Fast Electrical Modulation of Freely Propagating Terahertz Waves," *Appl. Phys. Lett.*, **93**, 2008, art. 091117.
39. W. J. Padilla, A. J. Taylor, C. Highstrete, M. Lee and R. D. Averitt, "Dynamical Electric and Magnetic Metamaterial Response at Terahertz Frequencies," *Phys. Rev. Lett.*, **96**, 2006, art.107401.
40. H.-T. Chen, W. J. Padilla, J. M. O. Zide, S. R. Bank, A. C. Gossard, A. J. Taylor and R. D. Averitt, "Ultrafast Optical Switching of Terahertz Metamaterials Fabricated on ErAs/GaAs Nanoisland Superlattices," *Opt. Lett.*, **32**, 12, 2007, pp. 1620-1622.
41. H.-T. Chen, W. J. Padilla, J. M. O. Zide, A. C. Gossard, A. J. Taylor and R. D. Averitt, "Active Terahertz Metamaterial Devices," *Nature*, **444**, 2006, pp. 597-600.
42. H.-T. Chen, W. J. Padilla, M. J. Cich, A. K. Azad, R. D. Averitt and A. J. Taylor, "A Metamaterial Solid-State Terahertz Phase Modulator," *Nature Photonics*, **3**, 2009, pp. 148-151.
43. H.-T. Chen, J. F. O'Hara, A. K. Azad, A. J. Taylor, R. D. Averitt, D. B. Shrekenhamer and W. J. Padilla, "Experimental Demonstration of Frequency-Agile Terahertz Metamaterials," *Nature Photonics*, **2**, 2008, pp. 295-298.
44. W. L. Chan, H.-T. Chen, A. J. Taylor, I. Brener, M. J. Cich and D. M. Mittleman, "A Spatial Light Modulator for Terahertz Beams," *Appl. Phys. Lett.*, **94**, 2009, art. 213511.
45. X. G. Peralta, I. Brener, W. J. Padilla, E. W. Young, A. J. Hoffman, M. J. Cich, R. D. Averitt, M. C. Wanke, J. B. Wright, H.-T. Chen, J. F. O'Hara, A. J. Taylor, J. Waldman, W. D. Goodhue, J. Li and J. Reno, "External Modulators for Terahertz Quantum-Cascade Lasers Based on Electrically-Driven Active Metamaterials," *Metamaterials*, **4**, 2, 2010, pp. 83-88.
46. H. Tao, A. C. Strikwerda, K. Fan, W. J. Padilla, X. Zhang and R. D. Averitt, "Reconfigurable Terahertz Metamaterials," *Phys. Rev. Lett.*, **103**, 2009, art. 147401.
47. J. Han and A. Lakhtakia, "Semiconductor Split-Ring Resonators for Thermally Tunable Terahertz Metamaterials," *J. Mod. Optics*, **56**, 2009, pp. 554-557.
48. T. Driscoll, S. Palit, M. M. Qazilbash, M. Brehm, F. Keilmann, B.-G. Chae, S.-J. Yun, H.-T. Kim, S. Y. Cho, N. M.

- Jokerst, D. R. Smith and D. N. Basov, "Dynamic Tuning of an Infrared Hybrid-Metamaterial Resonance Using Vanadium Dioxide," *Appl. Phys. Lett.*, **93**, 2008, art. 024101.
49. T. Driscoll, H.-T. Kim, B.-G. Chae, B.-J. Kim, Y.-W. Lee, N. M. Jokerst, S. Palit, D. R. Smith, M. Di Ventra and D. N. Basov, "Memory Metamaterials," *Science*, **325**, 2009, pp. 1518-1521.
50. T. Driscoll, G. O. Andreev, D. N. Basov, S. Palit, S. Y. Cho, N. M. Jokerst and D. R. Smith, "Tuned Permeability in Terahertz Split-Ring Resonators for Devices and Sensors," *Appl. Phys. Lett.*, **91**, 2007, art. 062511.
51. J. F. O'Hara, R. Singh, I. Brener, E. Smirnova, J. Han, A. J. Taylor and W. Zhang, "Thin-Film Sensing with Planar Terahertz Metamaterials: Sensitivity and Limitations," *Opt. Express*, **16**, 2, 2008, pp. 1786-1795.
52. S. Y. Chiam, R. Singh, J. Gu, J. Han, W. Zhang and A. A. Bettiol, "Increased Frequency Shifts in High Aspect Ratio Terahertz Split Ring Resonators," *Appl. Phys. Lett.*, **94**, 2009, art. 064102.
53. S. A. Tretyakov and S. I. Maslovski, "Thin Absorbing Structure for all Incident Angles Based on the Use of a High-Impedance Surface," *Microwave and Optical Technology Letters*, **38**, 3, 2003, pp. 175-178.
54. Y. Kotsuka, K. Murano, M. Amano and S. Sugiyama, "Novel Right-Handed Metamaterial Based on the Concept of 'Autonomous Control System of Living Cells,' and its Absorber Applications," *IEEE Trans. Electromag. Compat.*, **52**, 3, 2010, pp. 556-565.
55. Hu Tao, C. M. Bingham, A. C. Strikwerda, D. Pilon, C. Shrekenhamer, N. I. Landy, K. Fan, X. Zhang, W. J. Padilla and R. D. Averitt, "Highly Flexible Wide Angle of Incidence Terahertz Metamaterial Absorber: Design, Fabrication, and Characterization," *Phys. Rev. B*, **78**, 2008, art. 241103.
56. N. I. Landy, S. Sajuyigbe, J. J. Mock, D. R. Smith and W. J. Padilla, "Perfect Metamaterial Absorber," *Phys. Rev. Lett.*, **100**, 2008, art. 207402.
57. O. Luukkonen, F. Costa, C. R. Simovski, A. Monorchio and S. A. Tretyakov, "A Thin Electromagnetic Absorber for Wide Incidence Angles and Both Polarizations," *IEEE Transactions on Antennas and Propagation*, **AP-57**, 10, 2009, pp. 3119-3125.
58. F. Bilotti, A. Toscano, K. B. Alici, E. Ozbay, and L. Vegni, "Design of Miniaturized Narrowband Absorber Based on Resonant-Magnetic Inclusions," *IEEE Trans. Electromag. Compat.*, **53**, 1, 2011, pp. 63-72.
59. S. Sajuyigbe, M. Ross, P. Geren, S. A. Cummer, M. H. Tanielian and D. R. Smith, "Wide Angle Impedance Matching Metamaterials for Waveguide-Fed Phased-Array Antenna," *IET Microwaves, Antennas, and Propagation*, **4**, 8, 2010, pp. 1063-1072.
60. J. Gordon, C. L. Holloway and A. Dienstfrey, "A Physical Explanation of Angle-Independent Reflection and Transmission Properties of Metafilms/Metasurfaces," *IEEE Antennas Wireless Propag. Lett.*, **8**, 2009, pp. 1127-1130.
61. E. F. Kuester and C. L. Holloway, "A Low-Frequency Model for Wedge or Pyramid Absorber Arrays – I: Theory," *IEEE Trans. Electromag. Compat.*, **36**, 4, 1994, pp. 300-306.
62. J. Fang, "Generalized Perfectly Matched Layers for the Absorption of Propagating and Evanescent Waves in Lossless and Lossy Media," *IEEE Trans. Microwave Theory Tech.*, **44**, 12, 1996, pp. 2216-2222.
63. N. Engheta, "An Idea for Thin Subwavelength Cavity Resonators Using Metamaterials with Negative Permittivity and Permeability," *IEEE Antennas and Wireless Propagation Letters*, **1**, 2002, pp. 10-13.
64. T. Hand, S. Cummer, and N. Engheta, "The Measured Electric Field Spatial Distribution within a Metamaterial Sub-Wavelength Cavity Resonator," *IEEE Transactions on Antennas and Propagation*, **AP-55**, 6, 2007, pp. 1781-1788.
65. J. B. Pendry, D. Schurig and D. R. Smith, "Controlling Electromagnetic Fields," *Science*, **312**, 2006, pp. 1780-1782.
66. D. Schurig, J. J. Mock, B. J. Justice, S. A. Cummer, J. B. Pendry, A. F. Starr and D. R. Smith, "Metamaterial Electromagnetic Cloak at Microwave Frequencies," *Science*, **314**, 2006, pp. 977-980.
67. C. Enkrich, M. Wegener, S. Linden, S. Burger, L. Zschiedrich, F. Schmidt, J. F. Zhou, Th. Koschny and C. M. Soukoulis, "Magnetic Metamaterials at Telecommunications and Visible Frequencies," *Phys. Rev. Lett.*, **95**, 2005, art. 203901.
68. T. A. Klar, A. V. Kildishev, V. P. Drachev and V. M. Shalaev, "Negative-Index Metamaterials: Going Optical," *IEEE Journal of Selected Topics in Quantum Electronics*, **12**, 6, 2006, pp. 1106-1115.
69. A. V. Kildishev, W. Cai, K. Chettiar, H. Yuan, A. K. Sarychev, V. P. Drachev and V. M. Shalaev, "Negative Refractive Index in Optics of Metal-Dielectric Composites," *J. Opt. Soc. Am. B*, **23**, 3, 2006, pp. 423-433.
70. J. B. Pendry, "Negative Refraction Makes a Perfect Lens," *Phys. Rev. Lett.*, **85**, 2000, pp. 3966-3969.
71. A. Alù, A. Salandrino and N. Engheta, "Negative Effective Permeability and Left-Handed Materials at Optical Frequencies," *Opt. Exp.*, **14**, 4, 2006, pp. 1557-1567.
72. G. Dolling, C. Enkrich, M. Wegener, C. M. Soukoulis and S. Linden, "Low-Loss Negative-Index Metamaterials at Telecommunication Wavelengths," *Opt. Lett.*, **31**, 12, 2006, pp. 1800-1802.

73. T. J. Yen, W. J. Padilla, N. Fang, D. C. Vier, D. R. Smith, J. B. Pendry, D. N. Basov and X. Zhang, "Terahertz Magnetic Response from Artificial Materials," *Science*, **303**, 2004, pp. 1494-1496.
74. J. A. Gordon and R. W. Ziolkowski, "The Design and Simulated Performance of a Coated Nano-Particle Laser," *Optics Express*, **15**, 5, 2007, pp. 2622-2653
75. M. I. Stockman, "Spasers Explained," *Nature Photonics*, **2**, 2008, pp. 327-329.
76. J. A. Gordon and R. W. Ziolkowski, "CNP Optical Metamaterials," *Optics Express*, **16**, 9, 2008, pp. 6692-6716.
77. V. M. Shalaev, "Optical Negative-Index Metamaterials," *Nature Photonics*, **1**, 2007, pp. 41-48.
78. A. Alu and N. Engheta, "Theory of Linear Chains of Metamaterials/Plasmonic Particles as Subdiffraction Optical Nanotransmission Lines," *Physical Review B*, **74**, 2006, art. 205436.
79. A. Alu and N. Engheta, "Three-Dimensional Nanotransmission Lines at Optical Frequencies: A Recipe for Broadband Negative-Refractive Optical Metamaterials," *Physical Review B*, **75**, 2007, art. 024304.
80. N. Liu, H. Guo, L. Fu, S. Kaiser, H. Schweizer and H. Giessen, "Three-Dimensional Photonic Metamaterials at Optical Frequencies," *Nature Materials*, **7**, 2008, pp. 31-37.
81. J. Valentine, S. Zhang, T. Zentgraf, E. Ulin-Avila, D. A. Genov, G. Bartal and X. Zhang, "Three-Dimensional Optical Metamaterial with a Negative Refractive Index," *Nature*, **455**, 2008, pp. 376-379.
82. J. Gordon, C. L. Holloway and J. Booth, "Fluid Interactions with Metafilms/Metasurfaces for Tuning, Sensing, and Microwave Assisted Chemical Processes," *Physical Review B*, **83**, 2011, art. 205130.
83. S. Tretyakov, *Analytical Modeling in Applied Electromagnetics*, Norwood, MA, Artech House, 2003.
84. L. Brillouin, *Wave Propagation in Periodic Structures*, New York, Dover, 1953, Chapter 7.
85. A. Ishimaru, *Electromagnetic Wave Propagation, Radiation, and Scattering*, Englewood Cliffs, NJ, Prentice Hall, 1991, Chapter 7.
86. J. A. Kong, *Electromagnetic Wave Theory*, New York, John Wiley & Sons, 1986, Chapter 6.
87. T. K. Wu, *Frequency Selective Surface and Grid Array*, New York, John Wiley & Sons, 1995.
88. C. M. Soukoulis, *Photonic Band Gap Materials*, Dordrecht, The Netherlands, Kluwer Academic Publishers, 1996.
89. E. Sanchez-Palencia, *Non-Homogeneous Media and Vibration Theory*, Berlin, Springer-Verlag, 1980 (Lecture Notes in Physics No. 127).
90. G. Bouchitté and B. Schweizer, "Homogenization of Maxwell's Equations in a Split Ring Geometry," *Multiscale Modeling Simulation*, **8**, 3, 2010, pp. 717-750.
91. N. A. Khizhnyak, "Artificial Anisotropic Dielectrics: I, II and III," *Zh. Tekh. Fiz.*, **27**, pp. 2006-2013, 2014-2026 and 2027-2037, 1957 [in Russian]; English translation in *Sov. Phys. Tech. Phys.*, **2**, pp. 1858-1864, 1865-1876 and 1877-1886, 1957.
92. M. J. Dignam and M. Moskovits, "Optical Properties of Sub-Monolayer Molecular Films," *J. Chem. Soc. Faraday Trans. II*, **69**, 1973, pp. 56-64.
93. M. Johansson, C. L. Holloway, and E. F. Kuester, "Effective Electromagnetic Properties of Honeycomb Composites, and Hollow Pyramidal and Alternating Wedge Absorbers," *IEEE Transactions on Antennas and Propagation*, **AP-53**, 2, February 2005, pp. 728-736.
94. A. M. Nicolson and G. Ross, "Measurement of the Intrinsic Properties of Materials by Time Domain Techniques," *IEEE Trans. Instrum. Meas.*, **19**, 4, 1970, pp. 377-382.
95. W. B. Weir, "Automatic Measurements of Complex Dielectric Constant and Permeability at Microwave Frequencies," *Proc. IEEE*, **62**, 1, 1974, pp. 33-36.
96. D. R. Smith, S. Schultz, P. Markos and C. M. Soukoulis, "Determination of Effective Permittivity and Permeability of Metamaterials from Reflection and Transmission Coefficients," *Phys. Review B*, **65**, 2002, art. 195104.
97. R. W. Ziolkowski, "Designs, Fabrication, and Testing of Double Negative Metamaterials," *IEEE Transactions on Antennas and Propagation*, **AP-51**, 7, 2003, pp. 1516-1529.
98. X. Chen, T. M. Grezegorczyk, B.-I. Wu, J. Pacheco and J. A. Kong, "Robust Method to Retrieve the Constitutive Effective Parameters of Metamaterials," *Phys. Review E*, **70**, 2004, art. 016608.
99. S. Kim, E. F. Kuester, C. L. Holloway, A. D. Scher and J. Baker-Jarvis, "Boundary Effects on the Determination of Metamaterial Parameters from Normal Incidence Reflection and Transmission Measurements," *IEEE Transactions on Antennas and Propagation*, **AP-59**, 6, 2011, pp. 2226-2240.
100. C. L. Holloway, A. Dienstfrey, E. F. Kuester, J. F. O'Hara, A. K. Azad and A. J. Taylor, "A Discussion on the Interpretation and Characterization of Metafilms/Metasurfaces: The Two-Dimensional Equivalent of Metamaterials," *Metamaterials*, **3**, 2, 2009, pp. 100-112.

101. C. L. Holloway, E. F. Kuester, and A. Dienstfrey, "Characterizing Metafilms: The Connection Between Surface Susceptibilities and Effective Material Properties," *IEEE Antennas and Wireless Propagation Letters*, 2011.
102. L. A. Wainstein, "On the Electrodynamical Theory of Grids," in P. L. Kapitsa and L. A. Wainstein (eds.), *High-Power Electronics 2*, Oxford, UK, Pergamon Press, 1966, pp. 14-48.
103. C. L. Holloway, E. F. Kuester, and A. Dienstfrey, "On the Use of the Homogenization Technique to Determine a Generalized Boundary Conditions for Arbitrary Shaped Coated Wire Gratings," *Radio Science*, 2011 (in preparation).
104. T. B. A. Senior and J. L. Volakis, *Approximate Boundary Conditions in Electromagnetics*, London, UK, IEE, 1995.
105. M. I. Oksanen, S. A. Tretyakov and I. V. Lindell, "Vector Circuit Theory for Isotropic and Chiral Slabs," *J. Electromag. Waves Appl.*, **4**, 1990, pp. 613-643.
106. S. A. Tretyakov, A. J. Viitanen, S. I. Maslovski and I. E. Saarela, "Impedance Boundary Conditions for Regular Dense Arrays of Dipole Scatterers," *IEEE Transactions on Antennas and Propagation*, **AP-51**, 8, 2003, pp. 2073-2078.
107. O. Luukkainen, C. Simovski, G. Granet, G. Goussetis, D. Lioubtchenko, A. V. Räisänen and S. A. Tretyakov, "Simple and Accurate Analytical Model of Planar Grids and High-Impedance Surfaces Comprising Metal Strips or Patches," *IEEE Transactions on Antennas and Propagation*, **AP-56**, 6, 2008, pp. 1624-1632; "Correction," *IEEE Transactions on Antennas and Propagation*, **AP-58**, 2010, p. 2162.
108. A. D. Scher and E. F. Kuester, "Extracting the Bulk Effective Parameters of a Metamaterial via the Scattering from a Single Planar Array of Particles," *Metamaterials*, **3**, 1, 2009, pp. 44-55.
109. C. L. Holloway, D. Love, E. F. Kuester, J. Gordon, and D. A. Hill, "Use of GSTCs to Model Guided Waves on Metasurfaces/Metafilms," *IEEE Transactions on Antennas and Propagation*, submitted 2011.
110. J. Baker-Jarvis, M. D. Janezic, J. H. Grosvenor and R. G. Geyer, "Transmission/Reflection and Short-Circuit Line Methods for Measuring Permittivity and Permeability," NIST Technical Note 1355-R, NIST, US Department of Commerce, 1993.
111. B. Wathey, J. Tierney, P. Lidström and J. Westman, "The Impact of Microwave-Assisted Organic Chemistry on Drug Discovery," *Drug Discovery Today*, **7**, 6, 2002, pp. 373-380.
112. C. Oliver Kappe and D. Dallinger, "The Impact of Microwave Synthesis on Drug Discovery," *Nature Reviews Drug Discovery*, **5**, 2006, pp. 51-63.
113. C. Fermér, P. Nilsson, and M. Larhed, "Microwave-Assisted High-Speed PCR," *European Journal of Pharmaceutical Sciences*, **18**, 2, 2002, pp. 129-132.
114. R. W. Ziolkowski, P. Jin and C.-C. Lin, "Metamaterial-Inspired Engineering of Antennas," *Proc. IEEE*, **99**, 10, 2011, pp. 1720-1731.
115. P. Jin and R. W. Ziolkowski, "Metamaterial-Inspired, Electrically Small, Huygens Sources," *IEEE Antennas and Wireless Propagation Letters*, **9**, 2010, pp. 501-505.
116. P. Jin and R. W. Ziolkowski, "Multiband Extensions of the Electrically Small Metamaterial-Engineered Z Antenna," *IET Microwaves, Antennas & Propagation*, **4**, 8, 2010, pp. 1016-1025.
117. C.-C. Lin and R. W. Ziolkowski, "Dual-Band 3D Magnetic EZ Antenna," *Micr. Opt. Technol. Lett.*, **52**, 4, 2010, pp. 971-975.
118. P. Jin and R. W. Ziolkowski, "Broadband, Efficient, Electrically Small Metamaterial-Inspired Antennas Facilitated by Active Near-Field Resonant Parasitic Elements," *IEEE Transactions on Antennas and Propagation*, **AP-58**, 2, 2010, pp. 318-327.
119. P. Jin and R. W. Ziolkowski, "Low Q , Electrically Small, Efficient Near Field Resonant Parasitic Antennas," *IEEE Transactions on Antennas and Propagation*, **AP-57**, 9, 2009, pp. 2548-2563.
120. A. Erentok and R. W. Ziolkowski, "Metamaterial-Inspired Efficient Electrically-Small Antennas," *IEEE Transactions on Antennas and Propagation*, **AP-56**, 3, 2008, pp. 691-707.
121. R. W. Ziolkowski and A. Erentok, "Metamaterial-Based Efficient Electrically Small Antennas," *IEEE Transactions on Antennas and Propagation*, **AP-54**, 7, 2006, pp. 2113-2130.
122. A. Alu and N. Engheta, "Tuning the Scattering Response of Optical Nanoantennas with Nanocircuit Loads," *Nature Photonics*, **2**, May 2008, pp. 307-310.
123. I. A. Ibraheem and M. Koch, "Coplanar Waveguide Metamaterials: The Role of Bandwidth Modifying Slots," *Appl. Phys. Lett.*, **91**, 2007, art. 113517.
124. M. Quinten, A. Leitner, J. R. Krenn and F. R. Aussenegg, "Electromagnetic Energy Transport via Linear Chains of Silver Nanoparticles," *Opt. Lett.*, **23**, 17, 1998, pp. 1331-1333.
125. S. A. Tretyakov and A. J. Viitanen, "Line of Periodically Arranged Passive Dipole Scatterers," *Elec. Eng.*, **82**, 6, 2000, pp. 353-361.

126. M. L. Brongersma, J. W. Hartman and H. A. Atwater, "Electromagnetic Energy Transfer and Switching in Nanoparticle Chain Arrays Below the Diffraction Limit," *Phys. Rev. B*, **62**, 2000, pp. 16356-16359.

127. W. H. Weber and G. W. Ford, "Propagation of Optical Excitations by Dipolar Interactions in Metal Nanoparticle Chains," *Phys. Rev. B*, **70**, 2004, art. 125429.

128. R. A. Shore and A. D. Yaghjian, "Traveling Electromagnetic Waves on Linear Periodic Arrays of Lossless Spheres," *Electron. Lett.*, **41**, 10, 2005, pp. 578-580.

129. R. A. Shore and A. D. Yaghjian, "Traveling Electromagnetic Waves on Linear Periodic Arrays of Lossless Penetrable Spheres," *IEICE Trans. Commun.*, **E88-B**, 6, 2005, pp. 2346-2352.

130. M. Guasoni and C. De Angelis, "Analytical Approximations of the Dispersion Relation of a Linear Chain of Nanoparticles," *Opt. Commun.*, **284**, 7, 2011, pp. 1822-1827.

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